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Design of an Asynchronous Flash Analog-to-Digital Converter

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June 11, 2026

Resumo

A proliferação de dispositivos para a Internet das Coisas e de sensores biomédicos tem gerado uma crescente procura por interfaces de processamento de sinal de baixo consumo energético. Em muitas destas aplicações, os sinais de interesse são esparsos, caracterizando-se por longos períodos de inatividade. Os conversores analógico-digitais síncronos clássicos são inerentemente ineficientes nestes cenários, uma vez que dissipam potência dinâmica de forma contínua devido ao relógio global, independentemente da atividade do sinal de entrada.

Idealmente, a taxa de amostragem deveria ser adaptada dinamicamente às propriedades do sinal, por forma a minimizar o consumo energético associado à atividade do relógio. Contudo, a implementação de uma taxa de amostragem adaptativa não é direta. Neste contexto, a utilização de um conversor analógico-digital assíncrono surge como uma abordagem natural para processar sinais não uniformes. Os conversores assíncronos requerem alterações arquiteturais que permitam o seu funcionamento correto na ausência de um relógio global.

A adoção de controlo assíncrono em arquiteturas Flash é motivada, em primeiro lugar, pela necessidade de eliminar o consumo energético associado às redes de distribuição de relógio de alta velocidade, melhorando assim a eficiência energética do sistema. Para além disso, o funcionamento assíncrono oferece benefícios intrínsecos no tratamento da metaestabilidade dos comparadores e na redução da latência, tornando esta abordagem particularmente atrativa para aplicações orientadas a eventos, em que a atividade do sinal varia significativamente ao longo do tempo.

Esta dissertação propõe o projeto e a implementação de um conversor analógico-digital Flash assíncrono. Ao eliminar o relógio global, a arquitetura proposta alinha o consumo de potência com a atividade do sinal de entrada, atingindo teoricamente um consumo energético significativamente mais reduzido face às soluções síncronas convencionais.

Para dar resposta a este desafio, é proposta uma estratégia de calibração offline destinada a corrigir os desvios de offset dos comparadores, sem comprometer o funcionamento de alta velocidade característico da topologia Flash. O trabalho engloba a análise teórica, o projeto ao nível de esquemático e a validação do sistema através de co-simulação analógica e de sinal misto. O resultado é uma arquitetura de conversor sem relógio, robusta e eficiente em termos energéticos, que apresenta uma figura de mérito superior à das arquiteturas síncronas convencionais no contexto de aplicações com sinais esparsos.

Abstract

The proliferation of internet of things and biomedical sensors has created a high demand for low-power signal processing interfaces. In many of these applications, the signals of interest are sparse, characterized by long periods of inactivity. Classical synchronous analog to digital converters are inherently inefficient in such scenarios, as they consume dynamic power continuously due to the global clock, regardless of the input signal activity. Ideally the sampling rate should dynamically be adapted to the signal properties, so that power consumption due to the clock activity could be minimized. However, an adaptive sampling rate is not straightforward. As such, the use of an asynchronous ADC is a natural approach to sample non-uniform signals. Asynchronous ADCs require architectural changes, to allow proper operations without a global clock. The adoption of asynchronous control in Flash architectures is primarily motivated by the need to eliminate power consumption high-speed clock distribution networks, thereby improving energy efficiency. Furthermore, asynchronous operation offers intrinsic benefits in handling comparator metastability and reducing latency, making it particularly attractive for high-speed, event-driven applications where signal activity varies significantly over time. This dissertation proposes the design and implementation of an asynchronous Flash ADC. By removing the clock, the proposed architecture aligns power consumption with the input signal activity, theoretically achieving a much lower power consuption.

To address this, an offline trimming strategy is proposed to calibrate the comparator offsets without compromising the high-speed operation of the flash topology. The work encompasses the theoretical analysis, schematic design, and validation of the system through analog/mixed-signal co-simulation. The expected outcome is a robust, clockless ADC architecture that offers a superior figure of merit for sparse signal applications in compared to conventional synchronous architectures.

Keywords: Analog-to-Digital Converter, Asynchronous Design, Flash ADC, Level-Crossing Sampling, Offset Trimming, Low Power, IoT.

UN Sustainable Development Goals

The United Nations Sustainable Development Goals (SDGs) provide a global framework to achieve a better and more sustainable future for all. It includes 17 goals to address the world's most pressing challenges, including poverty, inequality, climate change, environmental degradation, peace, and justice.

Electronic systems play a pivotal role in modern society, acting as the backbone for smart infrastructure, healthcare monitoring, and environmental sensing. However, the massive deployment of battery-operated devices poses a significant challenge regarding energy consumption and electronic waste. This dissertation contributes directly to the efficiency and sustainability of these systems.

The specific Sustainable Development Goals addressed by this work are:

SDG 7 Affordable and Clean Energy: Ensure access to affordable, reliable, sustainable and modern energy for all. By optimizing the power consumption of data converters, we extend the battery life of devices and reduce the overall energy footprint of IoT networks.

SDG 9 Industry, Innovation and Infrastructure: Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation. This work advances the state-of-the-art in microelectronics by proposing novel asynchronous architectures that enable smarter and more efficient industrial sensors.

SDG	**Target**	**Contribution**	**Performance Indicators and Metrics**
7	7.3	By double the global rate of improvement in energy efficiency, this work reduces the power waste in standby modes of electronic sensors.	Reduction in Energy per Conversion (fJ/conv) compared to synchronous architectures.
9	9.5	Enhance scientific research, upgrade the technological capabilities of industrial sectors, encouraging innovation.	Successful validation of the proposed calibration algorithm and circuit topology.

"Our greatest glory is not in never falling, but in rising every time we fall"

Confucius

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List of Acronyms

IoT	Internet of Things
ADC	Analog-to-Digital Converter
DAC	Digital-to-Analog Converter
CMOS	Complementary Metal-Oxide-Semiconductor
LSB	Least Significant Bit
MSB	Most Significant Bit
SNDR	Signal-to-Noise-and-Distortion Ratio
SNR	Signal-to-Noise Ratio
SQNR	Signal-to-Quantization-Noise Ratio
ENOB	Effective Number of Bits
DNL	Differential Non-Linearity
INL	Integral Non-Linearity
BER	Bit Error Rate
FoM	Figure of Merit
SAR	Successive Approximation Register
LCS	Level-Crossing Sampling
GS/s	Giga-Samples per Second
MS/s	Mega-Samples per Second
SHA	Sample-and-Hold Amplifier
V_{th}	Threshold Voltage
V_{OS}	Input Offset Voltage
V_{DD}	Supply Voltage
INC	Increment event signal (upper level crossing)
DEC	Decrement event signal (lower level crossing)
DOUT	Digital output code
DOUT+	Upper digital threshold code
DOUT-	Lower digital threshold code
LC	Level Crossing
LFP	Local Field Potential
AP	Action Potential
RMSE	Root-Mean-Square Error

Chapter 1

Introduction

This chapter provides a brief introduction to the topic associated with the work carried out during the dissertation period. A contextualization is given, followed by a presentation of the research question and the main motivations for carrying out this study. The main objectives of this dissertation will also be outlined, as well as the methodology used to achieve them. Finally, the structure of the dissertation is presented.

We live in an era where smart devices are everywhere, known as the Internet of Things, IoT. While the processing and storage of information are inherently digital, benefiting from the scaling of Moore's Law, the physical world remains fundamentally analog. Physical variables such as temperature, sound and biological potentials are continuous in both time and amplitude. Consequently, the Analog-to-Digital Converter, ADC, serves as the critical interface bridging these two domains, enabling digital systems to interact with the real world [**johns_martin_2012**].

Energy efficiency has shifted from being a secondary performance metric to a primary design constraint, many of the sensors operate on limited battery or rely on energy harvesting, where every Joule of power dissipation has a direct effect on the system's autonomy. In this landscape, the data conversion block almost always takes place as the most spender in this field, especially in sensor interfaces where the digital transmission is duty-cycled.

However, a fundamental characteristic of many environmental and physiological signals is sparsity in between pulses. Signals such as electrocardiograms, voice activity, or environmental monitoring data are spread in bursts, they contain short periods of high information content followed by long intervals of inactivity.

Traditional data conversion approaches, based on uniform sampling and dictated by the Nyquist-Shannon theorem [**shannon_1949**], treat these silence periods with the same computational power and energetic waste as the active periods. This creates a paradigm of inefficiency, the system dissipates power to generate redundant digital samples that carry no new information. Recognizing and exploiting this sparsity is the key to unlocking ultra-low-power electronic interfaces.

Figure 1.1: Time-sparse signal representation and sampling strategies. (a) Time-sparse signal example showing burst activity with periods of inactivity. (b) Continuous-time event-driven sampling and conversion, which responds to signal changes. (c) Conventional uniform sampling operating at fixed Nyquist rate regardless of signal activity. (d) Discrete-time sampling with event-driven conversion, combining periodic and event-based approaches. This illustration demonstrates the efficiency of adaptive sampling methods for sparse signals compared to uniform sampling at the Nyquist rate [**lim_2013**].

The time-sparse nature of many signals can be observed in the burst waveform where information is concentrated in short intervals separated by long idle periods, as was demonstrated in 1.1. In particular, subfigures (b) and (d) highlight how continuous-time and discrete-time event-driven strategies avoid generating samples during inactivity, in contrast with conventional uniform sampling which keeps operating at a fixed Nyquist rate regardless of the signal content.

1.1 Motivation and Problem Statement

The energy waste in massive-scale IoT deployments and the inefficiency of conventional synchronous architectures when processing sparse data shows the need for a fundamental shift toward architectures that activate only when meaningful data events occur.

1.1.1 Inefficiency of Synchronous Sampling

Traditional Analog-to-Digital Converters operate under a fixed-rate sampling rate, dictated by a global clock signal (f_s). As established by the Nyquist-Shannon sampling theorem [**shannon_1949**], this rate is determined by the maximum possible bandwidth of the signal, $f_s \geq 2B$. However, in many real-world applications, the signal is often active for only a fraction of the time, meaning the true information content is much lower than the theoretical maximum bandwidth suggests.

This leads to a massive waste of energy due to the fundamental relationship governing dynamic power consumption in CMOS circuits [**baker_cmos_2010**]:

$$P_{dynamic} = \alpha \cdot f \cdot C_L \cdot V_{DD}^2 \tag{1.1}$$

where:

- $P_{dynamic}$ is the dynamic power dissipated.
- α is the activity factor (or switching factor).

- f is the clock frequency.
- C_L is the load capacitance being switched (e.g., in the clock tree).
- V_{DD} is the supply voltage.

In a uniformly clocked ADC, f is fixed by the Nyquist criterion and the clock network toggles every cycle, so α remains high even when the input signal is almost static. As a result, the converter continues to charge and discharge large clock and comparator capacitances at every period, dissipating dynamic power to generate redundant samples that do not carry new information. For example, if a sensor signal is active only 10% of the time but the ADC samples continuously, 90% of the switching events are spent tracking intervals where the input is essentially constant, directly illustrating how synchronous operation leads to massive energy waste for non synchronous signals.

1.2 Objectives and Scope of the Dissertation

Based on the limitations identified in traditional synchronous data conversion when dealing with asynchronous sensor signals, the primary objective of this work is to design and implement an energy-efficient ADC architecture, so this ADC must exploit signal inactivity to achieve a significant reduction in power consumption.

To achieve the overall goal of developing an efficient ADC, the following specific objectives are defined for this dissertation:

1. Asynchronous Architecture Design: To develop the circuit-level design of an asynchronous Flash ADC, minimizing dynamic power consumption by ensuring that power is only dissipated when an input signal event occurs.
2. Offline Trimming Implementation: To implement an offline trimming mechanism capable of detecting and compensating for the input offset voltage of the comparators.
3. Mixed-Signal Validation: To validate the complete system through analog and digital co-simulation. This validation must confirm the functionality and accuracy of both the asynchronous core and the trimming logic.
4. Performance Benchmarking: To quantify the energy efficiency of the proposed design using well known figures of merit. The results must be compared against the current state of the art synchronous ADCs and relevant asynchronous solutions.

In order to support these objectives, the state-of-the-art review is structured to progressively narrow the focus from general data-conversion theory to the specific asynchronous Flash architecture and trimming techniques addressed in this work. Chapter 2 begins by revisiting the fundamentals of sampling, quantization, and performance metrics for data converters, establishing the theoretical background required to interpret figures of merit such as SNDR, ENOB, and Walden FoM. It then surveys the main synchronous ADC architectures (Flash, SAR, Pipeline, and Sigma-Delta), highlighting their inherent power, speed and resolution trade-offs and identifying why conventional clocked Flash converters become inefficient when dealing with sparse signals.

1.3 Document Structure

This dissertation is organized into seven chapters. Chapter 1 introduces the topic, presents the motivation, problem statement, defines the objectives and scope of the work. Chapter 2 reviews the relevant background on analog to digital conversion, sampling paradigms, synchronous and asynchronous architectures, clockless event generation, calibration and trimming techniques, providing the basis for the proposed approach. Chapter 3 describes the design of the asynchronous Flash ADC, including the successive architectural choices and the final adopted solution. Chapter 4 presents the synchronous reference Flash ADC used for comparison. Chapter 5 discusses the trimming strategy adopted to correct comparator offset. Chapter 6 presents the simulation results and the performance evaluation of the proposed design. Finally, Chapter 7 summarizes the main conclusions and outlines directions for future work.

Chapter 2

Literature Review

Subsequent sections of the literature review focus on asynchronous and level-crossing sampling strategies, examining prior work on event-driven converters and their advantages for time-sparse signals, and finally discuss existing calibration and trimming techniques for comparator offset reduction, providing the reference framework against which the proposed bulk-driven resistive trimming mechanism will be compared.

This chapter presents the state-of-the-art in Analog-to-Digital Converters, reviewing theoretical foundations and analyzing survey data to justify the proposed architecture.

2.1 Fundamentals of Analog-to-Digital Conversion

This section provides the theoretical background required to understand the operation and performance characterization of data converters. It covers the fundamental steps of the conversion process such as sampling, quantization, and coding.

2.1.1 The Data Conversion Process

2.1.1.1 Ideal Data Conversion

The analog-to-digital conversion process acts as the bridge between the continuous physical world and the discrete digital domain. Conceptually, it involves two distinct operations: discretization in time, sampling, and discretization in amplitude, quantization. Ideally, this process should be instantaneous and lossless within the signal bandwidth of interest [**johns_martin_2012**].

2.1.1.2 The Sampling Operation

Sampling converts a continuous-time signal $x(t)$ into a discrete-time sequence $x[n]$ [**johns_martin_2012**].

According to the Nyquist-Shannon sampling theorem [**shannon_1949**], a band-limited signal with maximum frequency B can be perfectly reconstructed if the sampling frequency f_s satisfies:

$$f_s \geq 2B \tag{2.1}$$

Violating this condition results in aliasing, where high-frequency spectral components mixes into the baseband, becoming indistinguishable from the original signal.

2.1.1.3 The Quantization Operation

While sampling transforms continuous time in discrete values, quantization maps the continuous amplitude of each sample to one of a finite number of levels [**johns_martin_2012**]. An N-bit ADC divides the input range (V_{ref}) into 2^N discrete levels. The step size between adjacent levels is the Least Significant Bit (LSB):

$$V_{LSB} = \frac{V_{ref}}{2^N} \tag{2.2}$$

Unlike sampling, quantization is non-reversible and introduces a deterministic error. This error is typically modeled as white noise added, the quantization noise with a uniform probability distribution. For an ideal quantizer, the Signal-to-Quantization-Noise Ratio (SQNR) is given by the formula:

$$SQNR_{dB} \approx 6.02N + 1.76 \tag{2.3}$$

2.1.1.4 Coding

The final stage is encoding the quantized level into a binary format. The choice of coding scheme being the most common straight binary and gray code which depends on the system requirements for data processing and transmission, but does not affect the fundamental analog performance [**baker_cmos_2010**].

2.1.2 Classical Performance Metrics

To evaluate and compare different data converters objectively, a standard set of metrics is used. These are categorized into dynamic and static parameters.

2.1.2.1 Resolution and Sampling Rate

Resolution defines the theoretical dynamic range, while the sampling rate determines the maximum signal bandwidth. However, these are nominal values the actual performance is limited by noise and non-linearities [**johns_martin_2012**].

2.1.2.2 Signal-to-Noise-Distortion Ratio (SNDR)

Signal to noise ratio adding distortion ratio or SNDR is the primary dynamic metric. It is the ratio of the signal power to the total power of all noise and harmonic distortion components. From SNDR, the Effective Number of Bits, ENOB, is derived [**johns_martin_2012**]:

$$ENOB = \frac{SNDR_{dB} - 1.76}{6.02} \tag{2.4}$$

This value represents the true resolution of the converter at a specific input frequency.

2.1.2.3 Differential and Integral Non-Linearity (DNL and INL)

Static linearity is characterized by measuring the deviation of code transition levels from their ideal positions [**johns_martin_2012**]. The DNL measures the deviation of a single step width from the ideal 1 LSB. A DNL less than -1 LSB implies a missing code in the transfer function.

The INL is the cumulative sum of DNL errors, representing the deviation of the transfer curve from a straight line. Specific INL patterns like saw-tooth shapes can reveal systematic errors in the architecture, such as gain mismatch or non-linear biasing [**van_de_plassche_1994**].

2.1.2.4 Offset and Gain Error

These are linear errors, offset is a constant shift of the transfer characteristic as seen before, while gain error is a deviation in the slope [**johns_martin_2012**]. Unlike non-linearity, these errors preserve the signal shape and can often be calibrated out simply. Linear imperfections such as offset and gain error modify the global position and slope of the transfer characteristic without changing its shape, as shown in Fig. 2.1. In this representation, a constant DC shift moves the whole transfer curve up or down (offset error), while a slope deviation alters the full-scale span, the so called gain error, both of which can be compensated by simple digital calibration.

Figure 2.1: Adapted from [**matlab**] ADC transfer function showing offset and gain errors. Offset error represents a constant DC shift applied uniformly across all codes, while gain error modifies the slope of the transfer function. Both are linear errors that can be calibrated using digital post-processing techniques [**johns_martin_2012**].

2.1.2.5 Figures of Merit

Figures of merit (FoMs) provide normalized metrics to compare ADC performance across different architectures and technologies. The Walden FoM quantifies energy efficiency per conversion step:

$$\mathrm{FoM}_W = \frac{\mathrm{Power}}{2^{\mathrm{ENOB}} \cdot f_s} \tag{2.5}$$

where lower values indicate better performance.

The Schreier FoM extends this by incorporating bandwidth effectiveness for oversampled converters:

$$\mathrm{FoM}_S = \mathrm{SNDR} + 10\log_{10}\left(\frac{\mathrm{BW}}{\mathrm{Power}}\right) \tag{2.6}$$

with higher values representing superior performance.

2.1.3 State-of-the-Art Comparative Analysis

The analytical plots and tables presented below incorporate data from an exhaustive set of 140 state-of-the-art works. All entries were selected from the ISSCC and JSSC publications, due to their high prestige and stringent peer-review process in the field of data converters. Within these sources, only papers reporting both supply voltage (V_{DD}) and signal-to-noise-and-distortion ratio (SNDR) were retained, ensuring that all points in the performance envelopes are based on consistently documented electrical and dynamic performance metrics. To ensure transparency and reproducibility, the complete list of references used to generate these performance envelopes, covering Flash, SAR, Pipeline, and Sigma-Delta architectures, is provided in [**ALCoban_1999_113**, **AbhishekAgrawal_2023_100**, **AmyWhitcombe_2024_101**, **AnandSubramanian_2024_143**, **BGregoire_2008_** **BMalkietal_2012_47**, **Balmelli_2004_119**, **BenjaminHershberg_2019_30**, **BramVeraverbeke_2025_91**, **Breems_2007_130**, **CCLiu_2010_45**, **CHChan_2015_57**, **CHammerschmied_1997_39**, **CPochet_2022_110**, **CYLee_2022_140**, **CYLiouetal_2013_50**, **ChengChungHsu_2007_24**, **ChengEnHsieh_2025_93**, **ChiYunStanleyWang_2019_138**, **ChinYuLin_2016_61**, **Christen_2007_131**, **ChunChengLiu_2016_60**, **Dezzani_2003_116**, **Dorrer_2005_124**, **EJanssenetal_2013_51**, **EwoutMartens_2024_96**, **FMYaul_2014_54**, **FMichel_2011_134**, **FengChen_2003_115**, **Fontaine_2005_125**, **Fujimoto_2006_129**, **Gaggl_2004_121**, **Geelen_2006_19**, **GeunhaKim_2023_77**, **GianHoogzaadandRafRoovers_1999_6**, **Goes_2006_128**, **GovertGeelenandEdwardPaulus_2004_7**, **HHBoo_2015_29**, **HKHong_2015_58**, **HWei_2011_46**, **HZhao_2022_68**, **HanZhao_2025_87**, **HendrikvanderPloegandRobertRemmers_1999_105**, **Hesener_2007_4** **HongshuaiZhang_2023_71**, **HongzhiZhao_2023_73**, **Huber_2007_108**, **HwiCheolKim_2005_18**, **JCWang_2022_66**, **JLiu_2022_98**, **JaewonLee_2025_102**, **JiaChingWang_2023_97**, **JiangYu_2005_122**, **JianxiongXu_2025_1**, **JihangGao_2025_88**, **JinseokKoh_2005_123**, **JongmiLee_2024_144**, **JosephInginoJra** **JunhuaShen_2025_94**, **JunlinZhong_2025_103**, **JunyanHao_2023_32**, **KBult_1997_4**, **KaiChengCheng_202** **KentaroYoshioka_2017_62**, **KyeongwonJeong_2025_86**, **KyojinDChoo_2016_59**, **LiangQi_2019_137**, **MDing_2015_56**, **MFlynn_1998_5**, **MYoshioka_2010_44**, **MZhan_2022_67**, **ManxinLi_2023_70**, **MingleiZhang_2019_63**, **MingtaoZhan_2023_75**, **MingyangGu_2025_36**, **Mitteregger_2006_126**, **Ong_1997_112**, **PHarpe_2014_53**, **PHarpe_2015_55**, **PHarpeetal_2012_48**, **PHarpeetal_2013_49**,

PSchvan_2006_3, **PSchvan_2008_42**, **PatrickVogelmann_2018_135**, **QLiu_2022_141**, **RKapustaetal_2013_5** **RaresBodnar_2024_83**, **Ray_2006_20**, **SUKwak_1997_8**, **SandeepSanthoshKumar_2023_74**, **SangheonLee_2024_78**, **Senderowicz_1997_111**, **SerdarAbdullahYonar_2023_99**, **SeungChulLee_2007_23**, **SewonLee_2024_84**, **SharvilPatil_2024_35**, **ShengJuiHuang_2009_133**, **Shimizu_2006_107**, **SiddharthDevarajan_2009_26**, **Silva_2006_127**, **SiyuanYe_2024_95**, **SiyuanYe_2025_89**, **SungEnHsieh_2023** **TKang_2022_142**, **TWang_2022_69**, **Tabatabaei_2003_118**, **Verma_2006_40**, **Vleugels_2001_114**, **WLiu_2010_43**, **Waltari_2002_12**, **Wang_2005_16**, **XiyuHe_2024_80**, **YCHuang_2010_27**, **YChae_2008_132**, **YChaietal_2012_28**, **YHu_2022_139**, **YSegal_2022_65**, **YShimizu_2008_109**, **Yan_2003_117**, **YanboZhang_2023_72**, **YannanZhang_2025_92**, **Yao_2004_120**, **YaohuiLuan_2025_145**, **YiHuWang_2023_76**, **YingZuLin_2019_64**, **YongLim_2024_79**, **Yoshioka_2005_17**, **Yoshioka_2007_21**, **YoungDeukJeon_2007_22**, **YueCao_2025_37**, **YuefengCao_2023_33**, **YuefengCao_2024_34**, **YukoTambaandKazuoYamakido_1999_2**, **YunsongTao_2024_85**, **YunsongTao_2025_38**, **ZhuoyiChen_2024_** **ZhuoyiChen_2025_90**, **ZijianLiu_2025_104**].

The architecture marked with an [1] is a Flash ADC representing the best-case performance for it respective plots among all surveyed works. These highlight the performance envelope limits achieved by Flash converters in each figure. They serve as benchmarks for comparing other architectures like SAR, Pipeline, and Sigma-Delta.

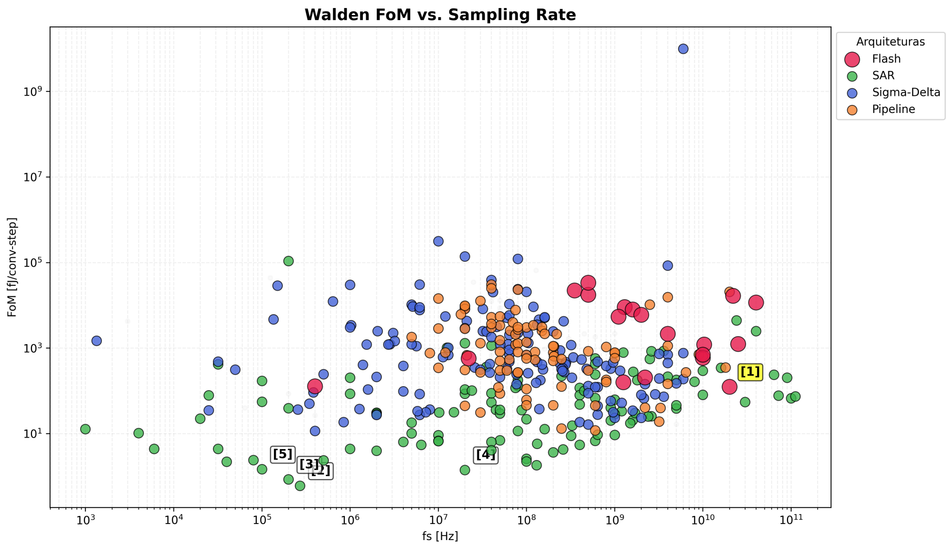

Figure 2.2: Walden FoM vs. Sampling Frequency (Flash reference [1] highlighted)

The evolution of the Walden Figure of Merit as a function of sampling rate for different ADC architectures is summarized in Fig. 2.2. This plot allows identifying the performance envelope across a wide range of f_s and highlights how specific implementations approach the energy-efficiency limit at their target bandwidth.

Table 2.1: Walden FoM vs. Sampling Frequency (Flash reference [1] highlighted)

Ref.	**Architecture**	**Publication**	f_s **[Hz]**
[1]	**Flash**	**Lukas [Lukas_2014]**	**20G**
[2]	SAR	Jang [**Jang_2018**]	270k
[3]	SAR	Ginsburg [**Ginsburg_2014**]	200k
[4]	SAR	Cano [**De_Cano_Garcia_2019**]	20M
[5]	SAR	Muller [**Muller_2015**]	200

2.1.4 Metrics Adaptation for Asynchronous Systems

The rigorous performance characterisation of asynchronous, event-driven analog-to-digital conversion systems requires a fundamental reassessment of the figures of merit that are conventionally applied to synchronous, uniformly sampled converters. In architectures based on level-crossing sampling, whether operating in continuous time or discrete time, the sampling rate is neither a static constant nor a predetermined design parameter. Instead, it is governed dynamically by the activity, slope, and rate of change of the analog input signal itself. As a direct consequence, the straightforward application of conventional formulations rooted in a fixed Nyquist sampling frequency, f_s, becomes mathematically inappropriate and fails to accurately capture the true efficiency of these topologies.

In conventional synchronous converters, the Walden Figure of Merit assumes a linear, time-invariant sampling frequency. However, as demonstrated by Weltin-Wu and Tsividis, level-crossing quantisers operate without a fixed global clock, generating new samples only when the analog stimulus crosses one of the discrete quantisation thresholds. This signal-activity-dependent sampling drastically reduces the production of redundant samples during intervals where the input behaves sparsely, extending the energy savings throughout the entire downstream digital processing chain. Furthermore, the absence of a uniform sampling clock alters the noise distribution across the frequency spectrum, yielding an output that is free from aliasing and from the conventional quantisation noise floor. This property allows level-crossing converters to achieve Signal-to-Noise-and-Distortion Ratios (SNDR) that can, under certain signal conditions, exceed the theoretical limits of conventional converters with the same nominal number of bits.

To model and fairly evaluate these asynchronous systems, it becomes necessary to adapt the Walden equation by replacing the fixed sampling frequency, f_s, with a metric based on the average rate of actual conversion events generated per unit time, denoted as f_{avg}, and expressed in effective conversion steps per second. The need for this methodological adaptation is clearly evident in the work of Mou *et al.*, which characterises the performance of a process-, voltage-, and temperature-robust discrete-time level-crossing ADC. Since the power consumption of the circuit fluctuates substantially with the dynamics of the input signal, ranging from 95 nW for highly time-sparse stimuli such as neural signals to 250 nW for generic waveforms, the efficiency of the device cannot be meaningfully quantified by means of a single static metric.

For this reason, the adapted Walden Figure of Merit is expressed as a function of the actual signal activity rate, formally adopting the unit of femtojoules per dynamic conversion step. Under this adapted methodology, energy efficiency is no longer reported as a fixed number but rather as a dynamic range, which, in the prototype of Mou *et al.*, spans values between 5.3 and 14.2 fJ per conversion step, depending strictly on the transition pattern induced by the input stimulus.

The physical basis for this variation in the Figure of Merit lies in the fact that the amplitude of signal change between successive samples, ΔV_{in}, determines the internal computational effort demanded from the converter. For slow events, where the variation falls between one and two LSB, only a simple linear update of ± 1 LSB is required in the feedback loop, minimising the number of comparisons performed during the active phase. Conversely, abrupt variations exceeding two LSB trigger adaptive full binary-search routines analogous to those employed in SAR converters, which locally increase both the number of comparisons per sample and the energy consumed per event.

In summary, the correct academic validation of asynchronous level-crossing architectures requires moving beyond the rigid assumptions of synchronous design. Indexing the Figure of Merit to the effective average activity rate, f_{avg}, ensures that signal sparsity is properly accounted for as a genuine factor of energy-efficiency gain, rather than being obscured by uniform sampling assumptions that do not reflect the actual operating conditions of event-driven converters.

2.2 Sampling Paradigms: Synchronous and Asynchronous

Data conversion systems traditionally rely on periodic temporal discretization to capture analog information. However alternative paradigms explore data acquisition mechanisms dictated directly by the behavioral dynamics of the input signal establishing a clear distinction between synchronous and asynchronous operations.

2.2.1 Non Uniform Sampling and Event Driven Systems

Non uniform sampling architectures eliminate the constraints imposed by a periodic global clock. In these event driven configurations data conversion is governed by amplitude variations in the spatial domain rather than fixed temporal intervals. Level crossing analog digital converters exemplify this methodology by generating a discrete sample exclusively when the analog input waveform intersects predefined quantization thresholds. Consequently the effective sampling rate of the system fluctuates dynamically in perfect correspondence with the instantaneous slope of the input signal. As demonstrated by Weltin Wu and Tsividis [**weltin2013**] this framework allows the sample generation rate to be proportional to the activity of the analog input establishing a system with activity dependent power consumption. The active components remain in a quiescent state during periods of low signal activity and dissipate dynamic power only when an explicit conversion event is triggered. This approach provides an alias free representation of continuous time signals mitigating the classic constraints of spectral folding [**weltin2013**]. In discrete time level crossing architectures such as the implementation by Mou et al. [**mou2026**] this flexibility is further enhanced through adaptive search schemes that switch between linear and binary tracking depending on whether a slow or fast level crossing event is detected based on comparator delay classification.

2.2.2 Uniform Sampling and Nyquist Rate Systems

Uniform sampling frameworks operate under the strict governance of the classical Nyquist Shannon theorem which mandates a constant sampling frequency equivalent to at least twice the highest frequency component of the target signal. Within these traditional architectures a rigid clock distribution network controls the timing of the sample and hold circuits alongside the subsequent quantization stages. This uniform methodology ensures a predictable data throughput and simplifies the implementation of digital signal processing algorithms due to the equally spaced temporal grid. However this mechanism forces the analog digital converter to continuously expend dynamic power irrespective of whether the input signal contains relevant information or remains stationary. This rigid temporal sampling creates a constant energetic overhead that limits efficiency when applied to signals with high temporal variations or prolonged periods of inactivity where the system continues to sample redundant data without useful informative content.

2.2.3 Signal Sparsity and Theoretical Energy Efficiency Benefits

Signal sparsity is an intrinsic property of many physical phenomena particularly within the domain of biomedical sensing applications where physiological waveforms exhibit long intervals of inactivity punctuated by sporadic bursts of high frequency information. In a conventional uniform sampling system the architecture must continuously operate at a high nominal Nyquist rate to prevent the loss of transient information during bursts which leads to substantial energy waste during silent periods. Event driven architectures directly exploit this inherent sparsity by adapting the instantaneous conversion rate to the information rate of the source. When the input signal remains static or confined within a specific inactive zone the conversion activity drops to zero minimizing dynamic switching losses. The theoretical energy efficiency benefit of this paradigm arises from this direct coupling between signal activity and hardware power dissipation enabling autonomous power scaling and optimized resource utilization as validated by the ultra low power consumption on the nanowatt scale reported by Weltin Wu and Tsividis [**weltin2013**] and Mou et al. [**mou2026**].

2.3 Synchronous Architectures

To address asynchronous ADCs we first need to address synchronous ADCs, they have a global clock signal that dictates sampling instances and synchronizes internal operations. While these architectures are highly mature and widely used, they face fundamental power efficiency trade-offs, particularly when operating at high frequencies or processing sparse data [**johns_martin_2012**].

2.3.1 Synchronous Flash ADC

The Flash ADC is conceptually the simplest and fastest architecture, performing a complete conversion in a single clock cycle [**baker_cmos_2010**]. It utilizes a resistive ladder to generate $2^N - 1$ reference voltages, which are compared simultaneously to the input signal by a large bank of comparators. This parallel comparison produces a thermometer code, that a digital logic block then converts into a standard binary output. Its primary limitation is the exponential increase in hardware complexity, area, and power consumption as resolution increases.

The thermometer term refers to the parallel output from the Flash ADC comparators, where '1's run up to the input voltage level like mercury in a thermometer, indicating signal strength before being converted to standard binary by an encoder.

2.3.1.1 The Power Bottleneck

The primary drawback of the synchronous Flash architecture is its exponential scaling. Survey data clearly illustrates that high speed translates into high power in these designs. For example, a 6-bit 500 MS/s CMOS Flash ADC reported in 1999 already consumed 400 mW [**Tamba**].

More extreme cases, such as a 22 GS/s 5-bit design, can consume up to 3W [**Schvan_2006**].

The continuous clocking of a massive comparator bank creates a significant energy dissipation even when the input signal is static. This lack of adaptivity makes synchronous Flash architectures increasingly unsuitable for battery-powered or energy-harvesting applications.

2.3.2 SAR (Successive Approximation Register)

The SAR ADC operates using a binary search algorithm [**van_de_plassche_1994**], for each sample the internal logic tests one bit at a time, starting from the most significant bit (MSB). A single comparator compares the input to the output of an internal digital to analog converter (DAC), if the input is higher, the bit is set to '1' otherwise, it is set to '0'. This process repeats for N cycles. Because it uses very few active components, it is the most energy efficient choice for low speeds.

The binary search operation of a SAR ADC is implemented by successively testing each bit through the interaction of the sample-and-hold amplifier (SHA), DAC, comparator, and control logic, as shown in Fig. 2.3. At every conversion step, the SAR logic updates the DAC output and the comparator decides whether the input is above or below the trial voltage, gradually converging from the MSB to the LSB towards the final digital code.

2.3.3 Pipeline

The Pipeline ADC breaks the conversion into several sequential stages [**johns_martin_2012**], each stage resolves a few bits of information, quantizes them, and passes the remaining error, the residue, to the next stage after amplification. This approach allows the ADC to work on multiple samples concurrently, obtaining very high throughput and high resolution, the problem relies in the inherent processing latency.

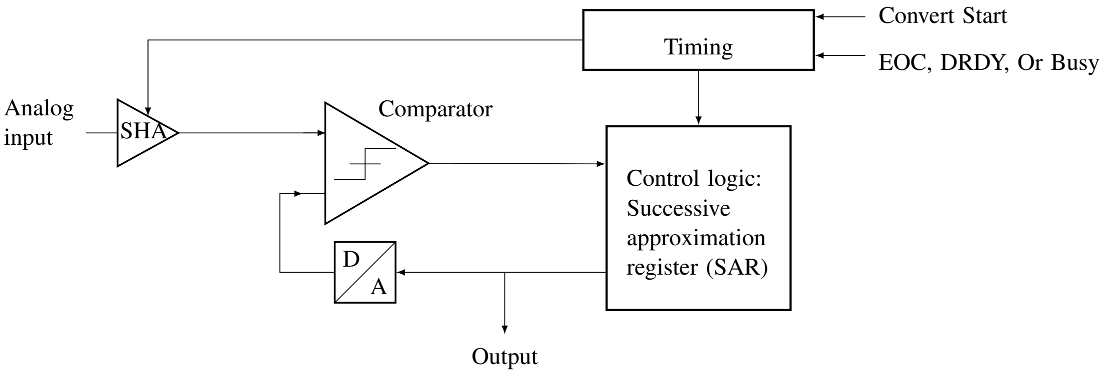

Figure 2.3: Successive Approximation Register (SAR) ADC architecture showing the sample-and-hold amplifier (SHA), comparator, digital-to-analog converter (DAC), and control logic. The SAR logic iteratively approximates the input signal by testing each bit from MSB to LSB, achieving high energy efficiency at moderate speeds [**van_de_plassche_1994**].

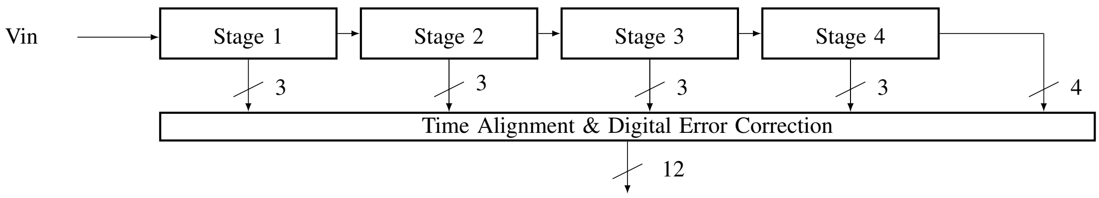

Figure 2.4: Pipeline ADC architecture showing multiple conversion stages operating in parallel on different input samples. Each stage resolves a subset of bits, generates a partial digital output, and amplifies the residue for the next stage. The pipelined structure enables high-throughput operation at the expense of increased latency and circuit complexity.

The pipelined converter processes multiple samples simultaneously by distributing the resolution across several cascaded stages, as shown in Fig. 2.4. Each stage resolves a subset of bits, generates a partial digital output, and amplifies the residue for the next stage, while a digital error-correction block aligns the timing of all stage outputs to produce the final high-throughput conversion result.

2.3.4 Sigma-Delta (ΣΔ)

The ΣΔ architecture relies on oversampling, sampling much faster than the Nyquist rate [**candy_1985**], and noise-modulation. A modulator integrates the difference between the input signal and a feedback version of the quantized output. This process pushes the quantization noise into higher frequencies, outside the signal band of interest, a digital filter then removes the high-frequency noise, resulting in high resolution and precision [**schreier_2005**].

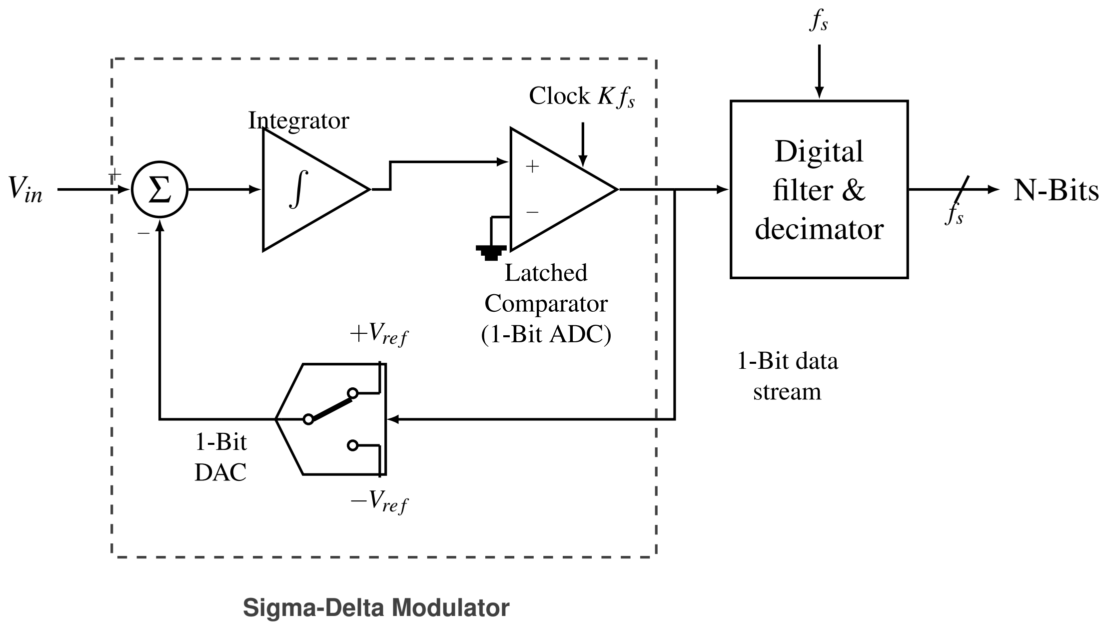

Figure 2.5: First-order sigma-delta ADC architecture illustrating the oversampling modulator with summer, integrator, 1-bit quantizer, feedback DAC, and digital decimation filter, which together implement noise shaping and high-resolution conversion [**candy_1985**, **schreier_2005**].

Oversampling and noise shaping are achieved by the feedback loop formed by summer, integrator, 1-bit quantizer, and digital decimation filter in the first-order sigma-delta architecture, as shown in Fig. 2.5. The modulator continuously integrates the difference between the analog input and the 1-bit DAC feedback, pushing most of the quantization noise to high frequencies, which are later removed by the digital filter.

2.3.5 Dual-slope

This integrating architecture [**microelectronics_1998**] measures the time required for a capacitor to charge and discharge, in the first phase, the capacitor is charged by the input voltage for a fixed period then for the second phase, it is discharged by a known reference voltage. The time it takes to return to zero is proportional to the average value of the input signal it's highly accurate and immune to high-frequency noise but is much slower than other architectures.

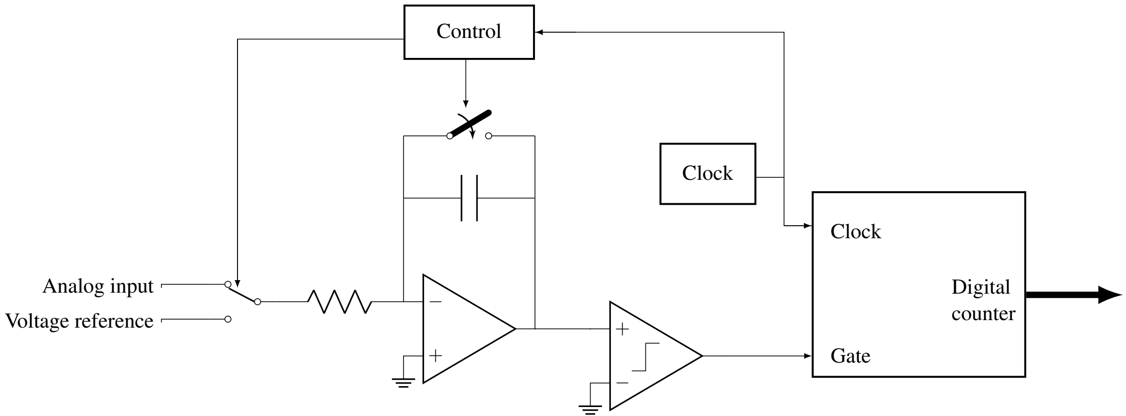

Figure 2.6: Dual-slope (integrating) ADC architecture showing the charging phase where an unknown input signal is integrated for a fixed time, and the discharge phase where a precise reference voltage drives the capacitor back to zero. The conversion time is proportional to the input voltage, providing excellent accuracy and noise rejection at the expense of low throughput.

The dual-slope architecture converts the input voltage into time by integrating it during a fixed interval and then discharging the capacitor with a known reference, as shown in Fig. 2.6. The digital counter measures the discharge interval controlled by the reference voltage, so the resulting count is proportional to the average value of the input signal and inherently rejects high-frequency noise.

2.4 Asynchronous Architectures

Asynchronous, or event-driven, architectures represent a fundamental paradigm shift in data conversion [**lim_2013**], unlike synchronous ADCs, which are bound by a global clock and the Nyquist-Shannon sampling theorem, asynchronous ADCs operate based on the signal's activity. This approach offers a more efficient alternative for processing sparse signals.

2.4.1 Level-Crossing Sampling (LCS)

The core principle behind many asynchronous ADCs is Level-Crossing Sampling (LCS) [**lim_2013**]. In traditional uniform sampling, the signal is captured at fixed time intervals (T_s), and the amplitude is quantized. In LCS, the process is inverted: the amplitude levels are fixed (quantization thresholds), and the ADC records the exact time at which the input signal crosses these thresholds.

This method is particularly powerful for signals that remain constant or change slowly over long periods. Instead of generating redundant samples that capture no new information, the LCS ADC remains idle, only producing a digital event when the signal effectively changes by more than the defined threshold.

Figure 2.7: Digital logic core of a level-crossing asynchronous ADC showing the two DACs that generate upper and lower thresholds around the input signal V_{IN}, the pair of comparators that produce the INC and DEC event signals, and the digital logic that updates the output code $DOUT$ and the time-stamp Δt based on each level crossing. The feedback paths from the digital logic to the DACs keep the thresholds centered around the most recent output level, enabling event-driven tracking of the input signal instead of uniform-rate sampling [**pae2022_lcadc**].

The core of the proposed event-driven conversion scheme is illustrated in Fig. 2.7. In this architecture, two on-chip DACs generate a pair of voltage thresholds placed symmetrically around the analog input signal V_{IN}, while two comparators continuously monitor the crossings of V_{IN} with respect to these levels. The upper comparator asserts the INC signal whenever V_{IN} rises above the upper threshold, whereas the lower comparator asserts the DEC signal when V_{IN} falls below the lower threshold.

Upon each level-crossing event, the digital logic block updates the encoded amplitude word $DOUT$ and the control words driving the DACs so that the thresholds are recentered around the new signal level. In addition, the logic measures the elapsed time since the previous event and outputs a time-stamp Δt, thereby representing the signal as a sequence of non-uniform events $(DOUT, \Delta t)$ instead of uniformly spaced samples. This mechanism concentrates conversion activity in intervals where the input varies rapidly and naturally suppresses conversions during flat regions, enabling an asynchronous ADC operation whose power consumption scales with the actual signal dynamics rather than with a fixed global sampling clock.

2.4.2 Flash ADC

The Asynchronous Flash ADC adapts the parallel structure of a standard Flash ADC but removes the sampling clock entirely. In this architecture, the comparators are not latched by a clock, they operate in continuous time, constantly monitoring the input voltage V_{in} against the reference ladder. The event generation happens when the input signal crosses a threshold, the corresponding comparator changes its output state immediately, this transition triggers an asynchronous logic to generate a digital output pulse.

2.5 Clockless Event Generation

2.5.1 The Voltage to Current Conversion Concept in Event Detection

A central challenge in the implementation of asynchronous Analog to Digital Converters is the generation of conversion events without relying on a global clock. Unlike synchronous architectures, where temporal discretization is externally imposed by periodic sampling, asynchronous systems require a mechanism capable of detecting meaningful changes in the input signal directly from its analog behavior. One efficient approach to achieve this objective is through voltage to current conversion.

The main idea behind voltage to current conversion is to translate variations of the input voltage into proportional current variations that can be continuously monitored by current domain circuits. Since current based processing often allows faster operation , this approach becomes particularly attractive for event driven architectures where response time and energy efficiency are critical.

In this context, the analog input signal is applied to a transconductance stage that converts the input voltage into a current proportional to its amplitude. This generated current is then compared against a reference current or a set of threshold currents that represent predefined quantization levels. Whenever the difference exceeds a decision threshold, an event is generated and propagated to the digital control logic.

This mechanism effectively transforms changes in signal amplitude into discrete temporal events. As a result, the converter remains inactive during periods of low signal variation and activates only when relevant transitions occur. Such behavior establishes a direct relationship between input activity and power consumption, making voltage to current conversion a fundamental building block for asynchronous data conversion systems.

2.5.2 Current Comparator Operating Principles

Current comparators are the fundamental decision elements used in event driven architectures operating in the current domain. Their purpose is to determine whether an input current is greater or smaller than a reference current and to generate a corresponding digital output that signals the occurrence of an event.

Unlike conventional voltage comparators, which rely on charging and discharging internal nodes until a voltage threshold is reached, current comparators process information directly through current differences. This characteristic enables reduced voltage excursions, lower parasitic charging requirements, and potentially faster response times.

In a typical implementation, two input branches receive the signal current and a reference current. The comparator evaluates the difference between these currents and amplifies the resulting imbalance until one of the output states becomes dominant. Depending on the adopted topology, this amplification may occur through current mirrors, regenerative feedback structures, resistive loads, or current conveyor stages.

The performance of a current comparator is usually evaluated according to several parameters including propagation delay, resolution, sensitivity, power consumption, and input impedance. In asynchronous event generation applications, propagation delay becomes particularly relevant since it directly determines the temporal precision of the generated conversion events.

An ideal current comparator should therefore combine low input impedance, fast regeneration, low static power consumption, and robust operation under process and temperature variations.

2.5.3 Topologies for Event Detection in the Literature

Several event detection topologies have been proposed in the literature to enable continuous monitoring of analog signals and asynchronous generation of digital events. These approaches mainly differ in the method used to compare currents and regenerate the output transition.

Current mirror based comparators represent one of the simplest solutions. Their operation relies on mirroring the input and reference currents and converting the resulting difference into a voltage decision. These structures offer reduced area and low implementation complexity but are generally limited by slower response due to large internal impedance nodes.

To overcome these limitations, regenerative comparators such as the Traff architecture introduce positive feedback mechanisms that accelerate the decision process and reduce propagation delay. These solutions improve speed but may introduce dead zones and increased sensitivity to operating conditions.

Alternative approaches employ complementary amplifier stages with resistive feedback, as proposed in continuous time current comparator architectures. By reducing internal voltage swings, these designs achieve a favorable compromise between power consumption and operating speed.

Other implementations incorporate hysteresis mechanisms to improve noise immunity and suppress unwanted switching around the decision threshold. Although this increases robustness, excessive hysteresis can degrade conversion accuracy.

More advanced solutions based on current conveyors and feedback enhanced amplifiers further improve sensitivity and dynamic response at the expense of increased circuit complexity.

The selection of an event detection topology therefore depends on the target trade off between speed, power consumption, implementation complexity, and robustness. For asynchronous Flash Analog to Digital Converters, the preferred solution should preserve continuous operation while minimizing propagation delay and avoiding unnecessary static power dissipation.

2.5.3.1 Current-Mirror-Based Current Comparator

The current-mirror-based comparator represents the simplest class of current-mode comparators. In this architecture, the input current and a reference current are applied to two branches of a mirror network that generates a difference current, which is then converted into a voltage across a high-resistance node. This voltage variation is sensed by a subsequent amplifier or inverter, producing a logic output that indicates whether the input is larger or smaller than the reference. The small-signal behaviour is governed by the output resistance of the mirror devices: a high intrinsic resistance allows small current differences to produce significant voltage variations, which improves sensitivity.

The main advantage of this structure is its simplicity, very small silicon area and low design effort, since it relies on basic current mirrors and standard logic inverters. It can also offer good static power characteristics if the bias currents are kept small, and the absence of complex feedback paths simplifies stability considerations. However, the high output resistance that improves sensitivity also leads to poor frequency response when driving capacitive loads, so the addition of inverters or buffers to obtain rail-to-rail output significantly degrades speed. Their propagation delay can vary strongly with process and temperature, making them usually less suitable for very high-speed asynchronous Flash ADCs unless carefully optimised.

2.5.3.2 Traff Current Comparator

The Traff comparator introduces a high-speed continuous-time current comparison scheme based on a source-follower input stage and a CMOS inverter with positive feedback. The input current is first converted into a voltage at a low-impedance node through a pair of source followers, this node then drives an inverting stage that regenerates the decision. The positive feedback loop keeps the input node from slewing rail-to-rail, so its voltage excursion is limited around a quiescent point and the effective input impedance remains low, which shortens the time required to develop the voltage difference that triggers the output transition. When the differential input current exceeds the resolution threshold, the latch-like action of the feedback rapidly pulls the output to one of the rails, providing a logic-compatible decision.

Among its main advantages, this architecture achieves a high speed, with reduced propagation delay for a given input current step and relatively low power consumption for high-speed operation. Furthermore, the absence of large high-resistance nodes improves the frequency response and makes the design attractive for high-bandwidth current-mode ADCs. On the other hand, the

Traff comparator exhibits a deadband region for very small input currents in which both input transistors operate near cut-off, temporarily increasing input resistance and degrading response time making it not suitable for this application.

2.5.3.3 Lu Chen Current Comparator

The continuous-time comparator proposed by Lu Chen employs a CMOS complementary amplifier with resistive feedback, followed by two resistive-load amplifiers and CMOS inverters. The core stage uses a pair of complementary transistors operating in saturation, with a MOS transistor in the linear region acting as a feedback resistor between input and output nodes. Small-signal analysis of this structure shows that, for typical device sizing, both input and output resistances of the complementary amplifier are approximately inversely proportional to the sum of the transconductances of the input devices, which strongly reduces the voltage swing at the internal nodes during a comparison. The following resistive-load amplifiers provide additional gain with relatively low static current, while the final inverters restore rail-to-rail logic levels.

This design achieves a favourable speed–power ratio because the reduced internal voltage swings translate into shorter response time for a given input current, while the resistive-load stages limit power consumption, particularly when the input current increases. Compared with other high-speed comparators, simulations reported by the authors show that the average delay for small currents is comparable to more complex solutions, but with lower or similar power, and without requiring external bias currents or voltages, improving process robustness. As a drawback, the achievable resolution remains dependent on the sizing of the feedback device and the effective transconductance of the input pair, so aggressive scaling of bias currents to save power can degrade sensitivity.

2.5.3.4 Hysteresis Current Comparator

The hysteresis current comparator studied by Ranjana Sridhar introduces a current gain stage with non-linear feedback and a CMOS inverter output. The gain stage consists of cascaded resistive amplifiers and an inverter enclosed by a pair of transistors that implement non-linear feedback, whose gates are driven by the output node while their sources sense the difference-node voltage. For small $|I_{\text{diff}}|$, both feedback transistors are off and the node behaves as a capacitive load around a quiescent operating point, whereas for larger positive or negative differences one of the feedback devices turns on, enhancing the loop gain around that point and accelerating the transition.

This non-linear feedback mechanism creates an effective hysteresis, once the output has switched, a finite current difference of opposite polarity is required to bring the internal node back to the decision boundary, which improves noise immunity and metastability behaviour. The use of multiple gain stages and feedback devices increases circuit complexity and area compared with simpler comparators, and the hysteretic behaviour must be carefully dimensioned so that the added memory effect does not introduce excessive offset or distort the transfer characteristic in applications where strictly monotonic behaviour is required.

2.5.3.5 Min and Kim Current Comparator

The Min and Kim comparator employs a resistive feedback network around a current-source inverting amplifier to reduce input and output impedances and improve dynamic response. The structure consists of several current-source amplifiers connected in cascade, with the first stage incorporating a resistor that feeds back a portion of the output voltage to the input node. In steady state, this feedback lowers the effective impedance seen at the input and output, thereby limiting the voltage swing required to represent a given current difference and shortening the time needed to reach a decision threshold at the amplifier output. A final CMOS inverter converts the amplified voltage into a digital logic level.

Simulation results reported in the literature indicate that this architecture achieves low propagation delay and good process immunity, with the resistive feedback helping to stabilise the operating point and reduce sensitivity to parameter variations. Additionally, the reduced node swings make the comparator suitable for relatively high-speed applications as long as the bias currents are chosen appropriately. A key limitation, however, is that both speed and resolution are strongly dependent on the bias current of the current-source amplifiers: achieving very high resolution requires higher bias currents, which directly increases static power dissipation and degrades energy efficiency. Moreover, because the design relies on current-source loads rather than resistive loads, its power consumption does not decrease when the input current increases, unlike other comparators where dynamic behaviour can partially trade off with static consumption.

2.5.3.6 CC-II-Based Current Comparator

The CC-II-based comparator proposed by Veepsa Bhatia uses a current subtractor followed by a current conveyor of the second generation and a multi-inverter output stage. The current subtractor is implemented with a p-type current mirror that produces the difference between the input current and a reference current across a resistor, effectively converting the difference into a proportional voltage. This voltage is then applied to the low-impedance X-terminal of a CC-II, whose behaviour is characterised by a unity voltage gain from Y to X and current conveyance from X to Z, so that the current difference is sensed at X and reproduced at the high-impedance Z terminal. The inverter chain at the output converts the conveyor's analog voltage into full-swing logic levels suitable for digital processing.

A key feature of this architecture is the use of positive feedback from one of the inverters back to the Y terminal of the conveyor, so that small variations in the X-node voltage induced by the input current difference are reinforced through the CC-II action. This positive feedback shortens the response time and increases sensitivity to small current steps, allowing the comparator to achieve sub-5 μA resolution with nanosecond-level propagation delay in simulations. The number of devices is also moderate, and the CC-II realisation provides inherently low input impedance at X and high output impedance at Z. On the downside, the presence of the op-amp or amplifier block used to realise the CC-II, together with the feedback network, increases design complexity and power compared with very simple mirror-based comparators. Furthermore, the performance is sensitive to the bandwidth and stability of the conveyor implementation, which must be carefully compensated to prevent oscillations due to the positive feedback.

2.6 Calibration and Trimming Techniques

In high-speed Flash ADCs, the accuracy of the system is fundamentally limited by the precision of the comparators. While the architecture fundamentals were established in the previous sections, practical implementations must address the non-idealities of the fabrication process, specifically the Input Offset Voltage (V_{os}).

2.6.1 The Component Mismatch Problem

In deep sub-micron CMOS technologies, transistors that are drawn with identical dimensions on the layout will exhibit slight differences in their electrical parameters after fabrication. This phenomenon, known as mismatch, affects the threshold voltage (V_{th}) and the current gain factor (β) of the differential pair in a comparator [**baker_cmos_2010**].

According to Pelgrom's Law cited in section 2.3.3 of [**johns_martin_2012**], the standard deviation of the threshold voltage mismatch (σ_{Vth}) is inversely proportional to the square root of the transistor area ($W \cdot L$):

$$\sigma_{V_{th}} = \frac{A_{V_{th}}}{\sqrt{W \cdot L}} \tag{2.7}$$

Where $A_{V_{th}}$ is a technology-dependent constant. This creates a critical trade-off, to minimize offset without calibration, transistors must be made large, which increases parasitic capacitance and degrades the ADC speed and power efficiency. Therefore, small transistors are used for speed, and calibration is needed to correct the resulting offset.

2.6.2 Calibration Classifications

Calibration techniques can be broadly categorized by their timing and domain [**baker_cmos_2010**]. In terms of timing there can be derived two types foreground calibration which interrupts normal operation to measure and correct errors and background calibration which operates continuously but adds significant complexity. In terms of domain there is digital calibration that corrects the output code mathematically, whereas analog calibration adjusts the circuit biasing or load conditions to nullify the offset at the source.

2.6.3 Trimming Techniques

2.6.3.1 Internal Loading

The internal resistive loading method involves placing a variable resistive network in parallel or in series with the output loads of the comparator stage. By digitally switching small resistors or MOS switches operating in the triode region in parallel with the load branch, the effective resistance R_L

is modulated. If the ADC has an offset, the trimming network is adjusted to introduce an equal and opposite value to effectively zero the error. This technique is typically implemented using a binary-weighted bank of PMOS transistors as the variable resistance, where the digital control code is determined at startup and stored in a register.

2.6.3.2 Reference Ladder Trimming

The reference ladder trimming technique corrects comparator offset by utilizing the main reference ladder as a calibration source [**yao_bulk_2011**]. In this architecture, a switching network selects specific voltages from the resistor ladder and applies them to the bulk terminals of the input transistors M1 and M2. This approach leverages the body effect to shift the threshold voltage (V_{th}) of the differential pair, thereby nullifying the input-referred offset caused by process mismatch.

As demonstrated in the simulation results of the threshold voltage ($|V_t|$) versus the ladder tap voltage (LT), the threshold voltage of transistors M1 and M2 increases linearly from approximately 356.5 mV to 372 mV as the calibration voltage is adjusted. In contrast, the threshold voltage of transistors M3 and M4 remains constant at approximately 356.5 mV because their bulk terminals are not connected to the tuning network. This targeted adjustment of the bulk potential for the input differential pair provides a tuning range sufficient to compensate for random process variations and ensure the accuracy of the comparator trip points.

Figure 2.8: Bulk-driven resistive reference ladder trimming architecture used for comparator offset calibration. The preamplifier input pair (M1, M2) has its bulk terminals driven by a selectable tap of the main reference resistor ladder via the offset calibration circuit (OSCAL). A digital control word selects one of the ladder taps (Rtap) through a tree decoder and multiplexers, generating a calibration voltage V_{cal} that modulates the bulk node V_B. By shifting the bulk potential of M1 and M2 while keeping the load devices at a fixed substrate potential, the threshold voltage of the input pair is adjusted through the body effect, providing a tunable range to cancel process-induced input offset in high-speed Flash ADC comparators [**yaobulk2011**].

Figure 2.9: Threshold voltage V_{th} variation of input differential pair transistors M1, M2 and load transistors M3, M4 as a function of the resistive ladder tap voltage LT. The graph demonstrates the body effect as the bulk potential of M1 and M2 is modulated via the reference ladder, providing a calibration range to compensate for process-induced mismatch [**yaobulk2011**].

The effectiveness of the bulk-driven trimming mechanism is illustrated by the controlled change of the input pair threshold voltage with the ladder tap voltage, as shown in Fig. 2.9 and Fig. 2.8. As the calibration voltage LT increases, the threshold of M1 and M2 shifts linearly while M3 and M4 remain fixed, providing the tuning range required to compensate comparator offset caused by random process variations [**yaobulk2011**].

2.6.4 Advantages of Resistive Trimming for Asynchronous ADCs

Compared to dynamic offset-cancellation techniques such as auto-zeroing, which rely on accurately timed clock phases ϕ_1 ϕ_2 and additional storage capacitors that must be periodically charged and discharged [**baker_cmos_2010**], resistive trimming is particularly well suited to an event-driven asynchronous architecture. Once the calibration bits are determined during a dedicated foreground calibration phase, the resistive trimming network remains static, meaning that no further switching activity or clocking is required and the comparator bias conditions are held purely by DC conduction paths. This static behavior ensures that the trimming circuitry does not disturb the intrinsic event-driven operation of the ADC, which should ideally remain idle whenever the input signal is not crossing a quantization threshold.

Another important advantage is that the resistive network can be connected so as to avoid placing large additional capacitances on the high-speed nodes of the comparator, thereby preserving the original small-signal bandwidth and regeneration speed. In contrast, dynamic calibration schemes typically introduce extra sampling capacitors and switch parasitics at sensitive nodes, which can degrade the comparator's response time and limit the maximum achievable sampling rate in high-speed Flash architectures [**baker_cmos_2010**].

2.7 Asynchronous vs. Synchronous

2.7.1 Advantage

The primary motivation for adopting asynchronous architectures is the direct relationship between signal activity and power consumption [**lim_2013**].

In a synchronous system, the clock tree and the comparators switch at every clock cycle, regardless of whether the input is changing, the power consumption is dominated by the fixed clock frequency [**baker_cmos_2010**]:

$$P_{sync} \approx f_{clk} \cdot C_{total} \cdot V_{DD}^2 \tag{2.8}$$

In an asynchronous ADC, the switching frequency is replaced by the event rate. If the signal is constant or slowly varying, the comparators and digital logic remain static, leading to the following proportionality:

$$P_{async} \propto Activity \times V_{DD}^2 \tag{2.9}$$

This property ensures that the energy consumed is always proportional to the information content of the signal which makes these ADCs significantly more efficient for sparse signal environments, where they can achieve near-zero power during periods of inactivity.

2.7.2 Comparative Evaluation

An analysis of state-of-the-art converters shows that asynchronous ADCs excel in energy efficiency across a wide range of sampling rates, particularly for low-to-medium bandwidths [**lim_2013**].

While Synchronous Flash ADCs are designed for peak performance at a specific frequency and suffer from a much higher power consumption caused by the continuous clocking, asynchronous designs scale their power consumption linearly with the input activity.

Table 2.2: Comparison of Power Consumption between Synchronous and Asynchronous Flash ADCs. Simulation results from ISSCC conference showing the power advantage of asynchronous operation for sparse signals.

Topology	**Average power (P [W])**
Synchronous	0.1016
Asynchronous	0.0314

The impact of this architectural difference on long-term technology trends is illustrated in Figure 2.10. The plot depicts the energy per conversion (P/f_s) over the last two decades. While both paradigms show a downward trend due to process node scaling, asynchronous designs consistently occupy the lower bound of the energy envelope. This suggests that eliminating the global clock is a key enabler for breaking the power efficiency barriers in modern designs.

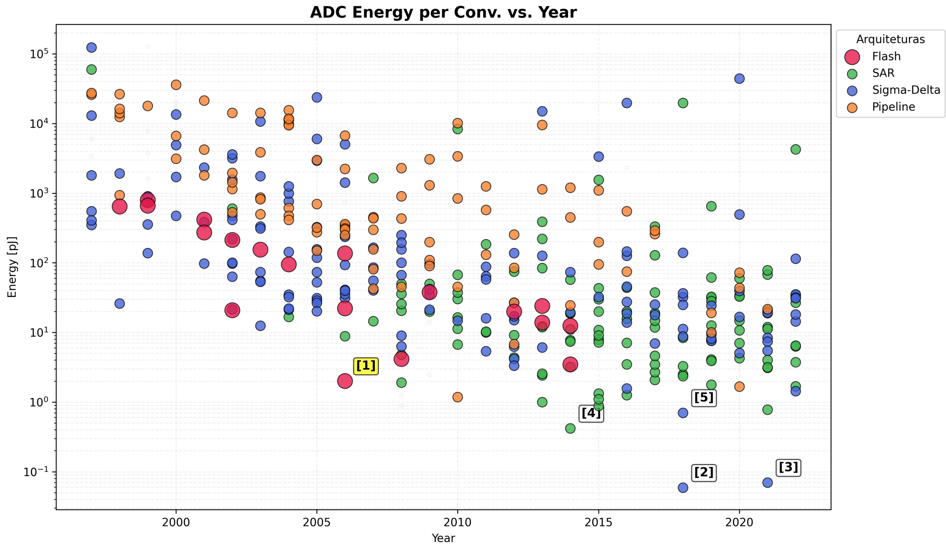

Figure 2.10: Energy per Conversion vs. Year (Flash reference [1] highlighted)

The long-term reduction in energy per conversion step driven by architectural innovation and technology scaling is depicted in Fig. 2.10. Each point in the plot corresponds to a published design, making it possible to compare how different ADC architectures fill the energy year plane and to identify the most efficient solutions. From the previous analysis, it becomes clear that

Table 2.3: Energy per Conversion vs. Year (Flash reference [1] highlighted)

Ref.	Architecture	Publication	Energy [pJ]
[1]	**Flash**	**Chen [Chen_2006]**	**2.02**
[2]	Sigma-Delta	Jiang [**Jiang_2018**]	0.059
[3]	Sigma-Delta	Veldhoven [**Veldhoven_2021**]	0.070
[4]	SAR	Ginsburg [**Ginsburg_2014**]	0.420
[5]	Sigma-Delta	Kim [**Kim_2018**]	0.703

synchronous architectures face a fundamental inefficiency when dealing with time-sparse sensor signals: the global clock forces continuous switching activity, so dynamic power remains proportional to the clock frequency even during long periods with no new information content. In contrast, event-driven schemes naturally decouple power consumption from the Nyquist rate and tie it instead to the actual rate of relevant signal transitions. Combining this observation with the survey data on state-of-the-art converters, where Flash ADCs define the speed frontier but suffer from poor Walden FoM at high sampling rates, while more energy-efficient SAR and sigma-delta architectures are optimized for lower bandwidths, motivates the choice of an asynchronous Flash topology as the core of this work. By removing the global sampling clock and operating comparators in a level-crossing, event-driven manner, the proposed asynchronous Flash ADC aims to preserve the intrinsic low-latency and high-speed advantages of the Flash architecture, while leveraging signal sparsity to significantly improve energy efficiency and close the FoM gap with the best-performing synchronous designs in the targeted operating region.

2.7.3 Chapter Summary and Design Directives

Asynchronous Flash ADCs combine a parallel comparison front-end with event-driven operation to exploit the time-sparse nature of many sensor signals. In these architectures, the input is processed as a current and the conversion is triggered only when the signal crosses predefined decision levels, avoiding unnecessary activity during idle periods. Compared with conventional synchronous converters, this enables a significant reduction in dynamic power, since large capacitive nodes associated with sampling clocks and comparator arrays are only switched when relevant information is present in the input waveform. In addition, operating in current mode relaxes voltage headroom constraints, facilitates low-voltage design and can offer improved speed for a given technology and power budget, making this approach well suited for energy-constrained IoT interfaces.

From an architectural point of view, the proposed converter follows a Flash topology, where a bank of comparators observes different reference levels in parallel and produces a thermometer-like code that is subsequently encoded into binary. As in a conventional Flash ADC, this guarantees single-cycle conversion and very low latency, but here the global clock is removed and the decision activity is instead driven by input events. This choice transfers the power bottleneck from the sampling network to the design of the current comparators themselves, since their intrinsic delay, input range and power consumption directly determine the achievable resolution, maximum event rate and overall energy efficiency. For this reason, a detailed study and comparison of several current comparator architectures was carried out before selecting the most suitable structure for the asynchronous core.

Chapter 3

Proposed Asynchronous Flash ADC Design

3.1 Architectural Evolution and Methodology

The design of the voltage comparator responsible for generating the digital output code of the analog-to-digital converter was not arrived at directly. It evolved through a sequence of candidate topologies, each evaluated against the constraints imposed by the target supply voltage, the required conversion accuracy, the effective input dynamic range, and the overall demands on circuit compactness. The following sections document this progression, presenting each topology that was considered and explaining the physical reasoning that either led to its adoption or prompted its replacement by a subsequent candidate.

3.2 Phase I: The Baseline Current-Mirror OTA Flash Configuration

3.2.1 Operating Principle and Core Topology

The current-mirror flash ADC (CMF-ADC) topology investigated in this phase is structured around two cascaded functional stages: a *transconductance cell*, which converts a differential input voltage into a proportional error current, and a *current-to-code converter*, which maps that current into a parallel thermometer-coded output without requiring a sampling clock. The complete circuit, annotated with the current-flow conventions used throughout this analysis, is shown in Figure **??**.

3.2.1.1 Transconductance Cell

The transconductance cell is a differential pair formed by transistors M_1 and M_2, both biased by a common tail current source of magnitude I_B. The non-inverting input of the pair receives the analog signal V_{in}, while the inverting input is driven by an adjustable reference voltage $V_{ref,c}$. An active current mirror, composed of diode-connected M_3 and mirroring transistor M_4, constitutes the load of the differential pair and steers the drain current of M_1 toward the output. An additional

NMOS mirror generates a third current branch I_3 as a replica of I_1, which is then distributed to the conversion stage.

Equilibrium operation. See Figure **??**(a) for the operation at equilibrium. When $V_{in} = V_{ref,c}$, M_1 and M_2 conduct equal quiescent currents of magnitude $I_B/2$. Because M_3 is diode-connected, its gate-to-source voltage adjusts to carry exactly $I_B/2$, and M_4, sharing this gate voltage, mirrors the same current to the output branch. The mirrored current from M_4 and the drain current of M_2 are therefore equal in magnitude, so their contributions cancel at the output node of the transconductance cell and the net error current is zero: $I_{err} = 0$. Under this condition, as indicated in Figure **??**(a), both branches carry identical currents in the same direction, resulting in no differential current available to drive the downstream converter.

Signal operation. See Figure **??**(b) for the differential signal analysis. When a differential voltage $v_{id} = V_{in} - V_{ref,c}$ is applied at the input, the symmetry of the pair is broken. Assuming matched transconductances $g_m = g_{m1} = g_{m2}$, the two transistors steer the tail current unequally, producing branch currents:

$$I_1 = \frac{I_B}{2} - I_{err}, \qquad I_2 = \frac{I_B}{2} + I_{err}, \tag{3.1}$$

where $I_{err} = g_m \cdot v_{id}/2$ is the error current. As annotated in Figure **??**(b), when $V_{in} > V_{ref,c}$, M_1 carries the excess current (indicated by the thickened arrow on the M_1 branch), while the current in the M_2 branch is correspondingly reduced (indicated by the crossed symbol). The mirror formed by M_3 and M_4 copies I_1 toward the output, and the resulting imbalance between this mirrored current and I_2 constitutes the differential output signal $2I_{err}$ that drives the code converter. The effective transconductance of the cell is therefore:

$$G_m = g_{m1} = g_{m2} = g_m. \tag{3.2}$$

3.2.1.2 Current-to-Code Converter and Threshold Generation

Rather than producing a single differential output voltage, the current-to-code converter distributes I_2 and I_3 across N parallel comparison nodes — one per thermometer output bit. Each node is loaded simultaneously by a PMOS current mirror, which replicates I_2 with a multiplication factor α_k, and an NMOS current mirror, which replicates I_3 with a complementary factor β_k. The two factors are chosen such that α_k decreases monotonically with the node index k while β_k increases correspondingly, as illustrated schematically in Figure **??**(c).

The binary output at node k is resolved by a simple current competition: if the PMOS sourcing current $\alpha_k I_2$ exceeds the NMOS sinking current $\beta_k I_3$, the node is pulled to a logic high, otherwise the NMOS pull-down dominates and the node settles to logic low. The quantisation threshold associated with node k is the specific input voltage at which the two competing currents are in equilibrium:

$$\alpha_k I_2 = \beta_k I_3 \Rightarrow v_{id}^{(k)} = \frac{I_B}{g_m} \cdot \frac{\alpha_k - \beta_k}{\alpha_k + \beta_k}. \tag{3.3}$$

Because the PMOS ratio α_k decreases monotonically with k, the thresholds $v_{id}^{(k)}$ also decrease monotonically, guaranteeing a valid thermometer code by construction. Nodes with a dominant PMOS mirror transition to high first as V_{in} rises above $V_{ref,c}$, followed progressively by nodes with higher NMOS ratios as the input continues to increase.

At the reference condition $V_{in} = V_{ref,c}$, the error current vanishes, $I_2 = I_3 = I_B/2$, and the state of each node reduces to comparing α_k against β_k directly. For an odd number of output bits with symmetric, evenly spaced ratio assignments, the central node satisfies $\alpha_k = \beta_k$ and is therefore at its decision boundary, producing the thermometer pattern `11···1?0···00` that is characteristic of a comparator array operating precisely at its reference level.

3.2.1.3 Adaptation to an Asynchronous Flash ADC

The circuit as described operates entirely in continuous time: no clock is needed to trigger a comparison, and the thermometer output updates immediately whenever V_{in} crosses one of the thresholds defined by Equation (3.3). This property is the primary motivation for adopting the CMF-ADC topology as the comparator front-end of an asynchronous Flash ADC.

In this context, each crossing event of V_{in} through a threshold level generates a spontaneous transition on the corresponding output bit without the intervention of any external timing reference. The ensemble of N output bits forms a thermometer code that tracks the input amplitude in real time: as V_{in} increases past each threshold, one additional bit transitions from low to high, and the reverse process occurs as V_{in} decreases. The asynchronous control logic described in Section **??** then detects these transitions and encodes them into the final binary output.

A further advantage in the ADC context is that monotonicity of the thermometer code is enforced at the topology level by the ordered mirror ratios, rather than relying on post-processing error correction. Bubble errors — isolated low codes within an otherwise high thermometer sequence — cannot arise as long as the PMOS ratios are strictly decreasing and the NMOS ratios are strictly increasing across the output nodes, since any such ordering guarantees that the threshold voltages $v_{id}^{(k)}$ form a strictly monotone sequence as required by (3.3).

The main design parameters that control the ADC's resolution and input range are the mirror ratios $\{\alpha_k, \beta_k\}$, the tail current I_B, and the transconductance g_m of the input pair. These relationships, along with the specific ratio assignments used in the implementation explored in this work, are discussed in Section **??**.

3.2.2 Current Comparator Topology Exploration and Characterization

The first topology investigated was the current-mirror flash ADC (CMF-ADC) introduced by Lee *et al.* [**lee2017**], originally proposed as the sensing front end of a coarse-fine dual-loop digital low-dropout regulator. Although the application context of that work differs substantially from the present design, the operating principle of the CMF-ADC constitutes a direct voltage-to-thermometer-code conversion mechanism that is directly applicable in the context of multi-bit comparison, making it a natural starting point for the architectural exploration.

The circuit is divided into two cascaded functional stages: a transconductance cell and a current-to-code converter. The transconductance cell is implemented as a differential pair. Its function is to translate the voltage difference between the input voltage, V_{in}, and an adjustable reference voltage, $V_{\text{ref,c}}$, into an error current I_{err}, yielding branch currents $I_1 = I_B - I_{\text{err}}$ and $I_2 = I_B + I_{\text{err}}$, where a positive I_{err} corresponds to V_{in} exceeding $V_{\text{ref,c}}$. A third current, I_3, replicates I_1 through a current mirror and is subsequently distributed to the conversion stage.

The current-to-code converter generates a multi-bit thermometer output by replicating I_2 and I_3 at each comparison node through PMOS and NMOS current mirrors, respectively, sized with deliberately asymmetric multiplication factors. In the five-bit implementation described in [**lee2017**], the PMOS mirror ratios decrease monotonically from $\times 14$ to $\times 6$ across the five output nodes specifically $\times 14$, $\times 12$, $\times 10$, $\times 8$, and $\times 6$ for bits $C_{\text{LPT}}[4]$ through $C_{\text{LPT}}[0]$ while the NMOS mirror ratios increase correspondingly from $\times 6$ to $\times 14$. At each comparison node, the output logic level is determined by whichever mirrored current dominates: when the PMOS contribution exceeds the NMOS contribution, the node is driven to a logic high, otherwise, the NMOS pull-down prevails and the output settles to a logic low.

To explain the operating principle clearly, it is easier to consider a configuration with an odd number of output bits, of which the five-bit case was the chosen one. An odd bit count provides a well-defined, symmetric midpoint whose comparison threshold aligns exactly with the condition $V_{\text{in}} = V_{\text{ref}}$, making the midpoint of the thermometer code unambiguous and facilitating a direct account of the circuit behaviour at the reference voltage. When V_{in} lies substantially below V_{ref}, the error current I_{err} is strongly negative, meaning $I_2 < I_B$ while $I_3 > I_B$. Since the NMOS mirrors replicate I_3 with larger multiplication factors than the PMOS mirrors replicate I_2 at every node, the NMOS current dominates throughout and the output code resolves to 00000.

As V_{in} rises toward V_{ref}, the magnitude of I_{err} decreases and the comparison nodes with the highest PMOS-to-NMOS mirror ratio are the first to transition. The node $C_{\text{LPT}}[4]$, carrying a PMOS mirror of ratio $\times 14$ against an NMOS mirror of only $\times 6$, presents the most favourable balance in favour of the PMOS side and therefore requires only a modest positive shift in I_2 relative to I_3 before its output transitions high. Node $C_{\text{LPT}}[3]$ ($\times 12$ PMOS, $\times 8$ NMOS) follows at a moderately larger error current, while the least significant nodes, $C_{\text{LPT}}[1]$ and $C_{\text{LPT}}[0]$, carry progressively more dominant NMOS mirrors and require a correspondingly larger current excess before their outputs transition.

At the exact condition $V_{\text{in}} = V_{\text{ref}}$, the error current vanishes and both I_2 and I_3 are equal to I_B. Nodes $C_{\text{LPT}}[4]$ and $C_{\text{LPT}}[3]$ deliver PMOS currents of $14I_B$ and $12I_B$, respectively, against NMOS contributions of $6I_B$ and $8I_B$, so both outputs are high. Nodes $C_{\text{LPT}}[1]$ and $C_{\text{LPT}}[0]$ exhibit the reverse relationship and remain low. The central node $C_{\text{LPT}}[2]$ carries equal PMOS and NMOS currents of $10I_B$ and is therefore in an exactly balanced. The resulting thermometer code at $V_{\text{in}} = V_{\text{ref,c}}$ is consequently 11?00, where the question mark denotes the indeterminate state of the central bit, fully consistent with the expected behaviour of a comparator operating precisely at its decision threshold.

The thresholds governing the transition of each bit are fully determined by the mirror ratios, as established analytically in [**lee2017**], and result in a monotonic, symmetric thermometer-coded response across the input voltage range. This architecture combines inherently fast conversion speed stemming from the absence of a conventional error amplifier, whose finite bandwidth would otherwise limit the response time of the control loop [**lee2017**] with a straightforward, hardware-efficient structure and an explicitly quantifiable input-output characteristic. It was therefore adopted as the first candidate for the present comparator design, and its operating principle informed all subsequent topological choices.

3.3 Phase II: The Complementary Dual-Input Flash Configuration

3.3.1 Motivation for Wide Input Common-Mode Range Extension

One of the primary objectives during the development of the proposed comparator architecture was to extend its input common-mode range as close as possible to the supply rails. Achieving a rail-to-rail input capability is desirable because it allows the converter to correctly process analog signals whose common-mode voltage may assume any value within the available supply range, from ground to V_{DD}, without requiring additional level-shifting or biasing circuitry.

This characteristic is particularly important in mixed-signal sensor interfaces, where the output voltage of the sensing element is often determined by the physical phenomenon being measured rather than by the requirements of the analog front-end. Restricting the admissible common-mode voltage consequently reduces the effective operating range of the converter and may require additional conditioning stages, increasing both power consumption and circuit complexity.

Furthermore, maximizing the input common-mode range enables better utilization of the available voltage swing. Since the signal occupies a larger fraction of the supply voltage, the effective dynamic range and signal-to-noise ratio can be improved without increasing the supply voltage itself. This consideration becomes especially relevant in modern low-voltage CMOS technologies, where the continuous reduction of V_{DD} inherently limits the available analog headroom.

A wide common-mode operating range also improves the robustness of the system against process, voltage, and temperature (PVT) variations. Small shifts in the input common-mode voltage caused by fabrication mismatch or supply fluctuations are less likely to drive the input transistors out of their intended operating region, thereby preserving the transconductance and ensuring predictable comparator behavior.

For these reasons, a complementary dual-input architecture capable of providing rail-to-rail operation was investigated as a second design candidate.

3.3.2 Rail-to-Rail Architecture Evaluation and Voltage Headroom Limitations

Although the proposed CMF-ADC satisfies the functional requirements of the event-driven conversion scheme, its input common-mode range remains inherently limited by the voltage headroom required by the differential input pair and the tail current source. This limitation becomes particularly critical in modern low-voltage CMOS technologies, where the continuous reduction of the supply voltage severely restricts the available analog signal swing. As a consequence, input signals whose common-mode voltage approaches either supply rail may drive the transconductance stage outside its intended operating region, degrading both linearity and sensitivity.

Extending the input common-mode range toward rail-to-rail operation is therefore highly desirable. Such capability allows the converter to process analog signals over the entire available supply range without requiring additional level-shifting or biasing circuitry, increasing compatibility with a wider variety of sensors and analog front-ends. Furthermore, maximizing the usable input voltage range improves the utilization of the available dynamic range, potentially leading to higher signal-to-noise ratio while maintaining the same supply voltage. It also increases robustness against process, voltage, and temperature (PVT) variations, since moderate shifts in the input common-mode voltage are less likely to move the input devices out of saturation.

Motivated by these considerations, a complementary dual-input architecture was investigated as a second design candidate. The topology employs two transconductance stages operating simultaneously: one based on an NMOS differential input pair and another based on a PMOS differential input pair. The rationale behind this arrangement is well established in rail-to-rail input amplifiers [**razavi**]. An NMOS differential pair operates efficiently when the common-mode voltage is close to the negative supply rail, where sufficient gate overdrive is available to maintain saturation, whereas a PMOS differential pair naturally provides coverage near the positive supply rail. By operating both stages simultaneously, at least one transconductor remains fully functional across the entire supply voltage range.

The implementation explored in this work did not simply combine the output currents of the two OTAs in parallel. Instead, one output branch of each OTA was connected to the PMOS current mirrors responsible for generating the individual bits of the thermometer code, preserving the current-comparison principle established in the original CMF-ADC architecture. The remaining output branch of each OTA was then cross-coupled to the complementary transconductor, such that the residual output current of one stage modulated the output node of the other, with the same arrangement applied symmetrically in the opposite direction.

The objective of this bidirectional cross-coupling was to exploit the complementary transconductance of both stages so that each OTA could reinforce the operation of the other near the boundaries of its valid input common-mode range. In principle, this mechanism would extend the effective dynamic operating region of the converter while preserving the event-driven current-comparison methodology.

Despite the conceptual advantages of this architecture, its practical implementation proved incompatible with the target supply voltage of $V_{DD} = 0.8$ V. A conventional five-transistor OTA already requires a minimum voltage headroom to accommodate the gate-to-source voltage of the differential pair, the saturation voltage of the tail current source, and the voltage drop across the active load. At a supply voltage of only 0.8 V, this headroom budget is already severely constrained.

The proposed dynamic-range extension mechanism required the insertion of one additional transistor in the critical signal path beyond the original five-transistor topology. Although seemingly a minor modification, every additional stacked transistor introduces an extra overdrive or saturation voltage requirement. Consequently, the cumulative voltage headroom exceeded the available supply, preventing all active devices from remaining in saturation over the intended operating range and therefore compromising the correct functionality of the architecture.

An attempt was made to recover the lost headroom by replacing the conventional current mirrors with the low-voltage current mirror topology proposed by Razavi [**razavi**]. This structure reduces the minimum voltage required across the mirror branch by eliminating one stacked gate-to-source voltage drop from the critical path, albeit at the expense of increased circuit complexity. Nevertheless, even with this modification, the combined voltage requirements of the differential input pair, the tail current source, the low-voltage current mirror, and the cross-coupled output stage still exceeded the available supply voltage.

No feasible bias point could be identified that simultaneously maintained all transistors within their intended operating regions while preserving the desired dynamic-range extension mechanism. Consequently, although the proposed rail-to-rail architecture was conceptually attractive, the stringent headroom constraints imposed by a 0.8 V supply rendered its practical implementation unfeasible. For this reason, the architecture was abandoned in favor of the simpler five-transistor OTA presented in the following section, which provided a significantly better compromise between functionality, voltage headroom, implementation complexity, and overall robustness.

3.4 Phase III: The Final Adopted Architecture

3.4.1 Five Transistor OTA Core and Optimized Transistor Sizing

Following the rejection of the complementary dual input approach, the design converged toward a classical five transistor operational transconductance amplifier as the comparator front end. This topology consists of a differential input pair loaded by an active current mirror implemented with two PMOS transistors, where one device operates in diode connection and sets the gate voltage of the mirrored branch. This structure forms the conventional active load of a differential pair and constitutes the defining element of the five transistor OTA architecture [**razavi**].

By eliminating the additional stacked devices required by the previous architecture, this solution satisfies the voltage headroom constraints imposed by the selected supply voltage of $V_{\mathrm{DD}} = 0.8$ V while maintaining correct operation over the intended input range.

The initial implementation evaluated the OTA under open loop conditions in order to maximize gain and improve sensitivity to small differential input voltages. Through transistor sizing optimization, high voltage gain could be achieved and therefore increase comparison sensitivity. However, practical simulations revealed a critical limitation of this approach. Small unintended differential inputs originating from offset, thermal noise, or marginal overdrive conditions caused the output node to rapidly saturate at one of the supply rails before the intended comparison could be completed.

Once saturation occurred, the recovery process became limited by the available bias current responsible for charging or discharging the high impedance output node. This significantly increased settling time and introduced variability in consecutive conversion cycles.

To overcome this limitation, the OTA was reconfigured as a unity gain voltage follower by directly connecting the output node to the inverting input and establishing full negative feedback. Under this configuration, the differential input voltage remains close to zero in steady state and the amplifier continuously operates in its linear region.

As a consequence, the OTA acts as a low impedance replica of the input signal, reproducing V_{in} at its output while avoiding saturation effects. The resulting accuracy is determined primarily by the open loop gain and the input referred offset of the amplifier.

3.4.2 Ratioed Output Slicers Design

The quantization stage of the converter is implemented through a bank of CMOS slicer inverters directly connected to the output of the voltage follower. Unlike the previously explored current mirror based architecture, where threshold positioning was determined by current scaling, the present design defines each comparison level through the PMOS to NMOS sizing ratio of each inverter.

The switching threshold of a CMOS inverter is determined by the relative strengths of the pull up and pull down networks and can be approximated by

$$V_M \approx \frac{V_{\text{DD}}\sqrt{r}}{1+\sqrt{r}} \tag{3.4}$$

where

$$r = \frac{(W/L)_p}{(W/L)_n}$$

represents the ratio between PMOS and NMOS transistor dimensions.

An increase in r shifts the switching threshold toward higher voltages, whereas lower values move the transition below the mid supply level. By assigning distinct transistor ratios to each inverter, a distributed set of decision thresholds is created along the input voltage axis.

Each slicer therefore acts as an independent comparator operating at a predefined threshold voltage. Collectively, the outputs generate a thermometer coded representation of the input signal.

For the implemented five bit configuration, the most significant output bit, out_4, uses a transistor ratio of $r = 1.5$, producing a switching threshold above 0.4 V. In contrast, the least significant bit, out_0, adopts $r = 0.5$, resulting in a lower threshold. The intermediate slicers are distributed between these limits to generate approximately symmetric threshold spacing around the reference level.

As the input voltage increases from ground toward V_{DD}, each inverter switches sequentially according to its assigned threshold. At $V_{in} = 0.4$ V, slicers with $r > 1$ remain in logic high state while slicers with $r < 1$ have already transitioned to logic low. The inverter designed with $r = 1$ operates exactly at its switching point and therefore enters a metastable region.

The resulting thermometer code becomes 11?00, reproducing the same behavior previously observed in the CMF ADC architecture.

3.4.3 Asynchronous Control Logic and Clockless Event Generation Network

A key characteristic of the proposed architecture is the complete removal of the global clock from the conversion process. Instead of forcing periodic operation, event generation occurs exclusively as a consequence of changes in the analog input signal.

The OTA configured as a voltage follower continuously tracks the input signal and distributes this low impedance representation to all slicers simultaneously. Each slicer independently evaluates whether the input has crossed its corresponding threshold and immediately updates its output state.

The collection of slicer outputs forms an asynchronous thermometer code whose transitions directly represent input activity. Whenever one or more thresholds are crossed, a digital event is naturally generated without requiring any external timing reference.

This event driven operation establishes a direct relationship between signal activity and switching activity inside the converter. During intervals where the input remains constant, the slicers remain static and dynamic power consumption approaches zero. Conversely, periods of rapid input variation automatically increase event generation frequency and conversion activity.

Compared with previous current mirror based implementations, this architecture introduces a clear separation between input buffering and threshold generation. The OTA is responsible only for reproducing the input voltage, while threshold definition is achieved entirely through transistor sizing in the slicer stage.

This separation simplifies the design procedure, improves robustness against saturation, and removes additional voltage headroom constraints, resulting in a compact and scalable architecture suitable for low voltage asynchronous Flash conversion.

3.5 Analytical Modeling and Design Constraints

In this section, the core theoretical boundaries and design trade-offs governing the asynchronous Flash Analog-to-Digital Converter (ADC) are established. Unlike conventional synchronous architectures, where performance limits are rigidly bound to a global clock frequency, the constraints of an asynchronous level-crossing or tracking framework are deeply intertwined with the instantaneous dynamics of the input signal, circuit propagation delays, and noise statistics.

3.5.1 Maximum Input Wave Frequency and Slope Overload

In synchronous ADCs, the maximum input signal frequency is strictly governed by the Nyquist-Shannon sampling theorem ($f_{in} < f_s/2$). In contrast, an asynchronous Flash or level-crossing ADC operates without a uniform sampling grid, its primary limitation is dictated by the maximum tracking velocity of its internal loops, commonly characterized as the Slew Rate (SR) constraint. A failure to track rapid signal variations leads to a severe distortion phenomenon known as slope overload.

To model this constraint analytically, let us consider a full-scale sinusoidal input voltage $V_{in}(t)$ defined as:

$$V_{in}(t) = \frac{V_{FS}}{2}\sin(2\pi f_{in}t) \tag{3.5}$$

where V_{FS} represents the full-scale voltage range of the converter and f_{in} is the input frequency. The time derivative of this signal yields its instantaneous slope, which achieves its maximum absolute value at the zero-crossing points:

$$SR_{max} = \left.\frac{dV_{in}(t)}{dt}\right|_{max} = \pi f_{in}V_{FS} \tag{3.6}$$

The asynchronous Flash ADC resolves the input signal by triggering a new state or quantization conversion whenever the input crosses a discrete threshold voltage step Δ, corresponding to the nominal resolution of N bits:

$$\Delta = \frac{V_{FS}}{2^N} \tag{3.7}$$

In an ideal, noiseless scenario, the asynchronous control logic and comparator array require a finite minimum processing time to complete a single conversion cycle—encompassing tracking, comparison, latching, and resetting. This aggregate time window is denoted as the total conversion propagation delay τ_{conv}. Consequently, the maximum rate at which the ADC can track voltage transitions (the converter's intrinsic Slew Rate, SR_{ADC}) is bounded by:

$$SR_{ADC} = \frac{\Delta}{\tau_{conv}} = \frac{V_{FS}}{2^N\tau_{conv}} \tag{3.8}$$

To circumvent slope overload distortion, the maximum slope of the input signal must never exceed the tracking capability of the asynchronous circuitry ($SR_{max} \leq SR_{ADC}$):

$$\pi f_{in}V_{FS} \leq \frac{V_{FS}}{2^N\tau_{conv}} \tag{3.9}$$

Rearranging this inequality yields the absolute maximum input frequency $f_{in,max}$ that the converter can faithfully digitize without undergoing slope overload under ideal conditions:

$$f_{in,max} = \frac{1}{\pi 2^N\tau_{conv}} \tag{3.10}$$

When transient noise is introduced into the simulation framework, the maximum operational

Slew Rate undergoes significant degradation. Superimposed thermal and substrate noise creates rapid, localized voltage fluctuations (dV_{noise}/dt). This high-frequency noise contribution artificially inflates the effective local slope of the signal, causing the voltage to bounce back and forth across a quantization threshold. This rapidly consumes the available processing window τ_{conv} and manifests as thermometer-code meta-stability or *bubble errors*. Consequently, the practical maximum Slew Rate under transient noise conditions is substantially lower than the ideal nominal value, causing slope overload to manifest at lower input frequencies and increasing the measured bubble rate.

3.5.2 Resolution and Power Dissipation

One of the most compelling advantages of asynchronous data converters is their signal-dependent power profile. In a standard synchronous Flash ADC, the power consumption remains largely constant because the entire comparator bank ($2^N - 1$ devices) and the clock distribution network are continuously toggled at the sampling frequency f_s, irrespective of the input signal activity. In an asynchronous implementation, power dissipation scales dynamically with the input signal's rate of change.

When the input signal is entirely static ($f_{in} = 0$), no quantization thresholds are crossed. Under this condition, the dynamic switching power ideally drops to zero, and the system only dissipates static power (P_{static}), which originates from the continuous bias currents in the analog references, pre-amplifiers, and comparators:

$$P_{total} = P_{static} + P_{dynamic} \tag{3.11}$$

For a dynamic input signal, such as the full-scale sinusoid at frequency f_{in}, the signal traverses the entire full-scale range V_{FS} twice within a single period $T = 1/f_{in}$ (two upward excursions and two downward excursions across the 2^N levels). Thus, the average number of quantization level crossings per second—which can be interpreted as an effective average sampling frequency $f_{s,avg}$—is proportional to both the frequency and the resolution:

$$f_{s,avg} = 2 \cdot 2^N \cdot f_{in} = 2^{N+1} f_{in} \tag{3.12}$$

Assuming that each individual asynchronous conversion cycle draws an average switching energy E_{conv} from the power supply V_{DD}, the dynamic power consumption $P_{dynamic}$ can be modeled analytically as:

$$P_{dynamic} = E_{conv} \cdot f_{s,avg} = E_{conv} \cdot 2^{N+1} f_{in} \tag{3.13}$$

Substituting this back into the total power equation yields:

$$P_{total} = P_{static} + E_{conv} \cdot 2^{N+1} f_{in} \tag{3.14}$$

Equation (3.14) explicitly demonstrates that while $P_{dynamic}$ retains a beneficial linear relationship with the signal frequency f_{in}, it scales exponentially with the nominal resolution N.

In practical circuit implementations, the physical performance is constrained by noise, which splits the design analysis into nominal resolution (N) and Effective Number of Bits ($ENOB$). The $ENOB$ is derived directly from the Signal-to-Noise-and-Distortion Ratio ($SNDR$ or $SINAD$):

$$ENOB = \frac{SNDR_{dB} - 1.76}{6.02} \tag{3.15}$$

When transient noise is factored in, the Signal-to-Noise Ratio (SNR) and $SNDR$ degrade, lowering the $ENOB$. Crucially, the presence of transient noise introduces spurious, non-monotonic threshold crossings (chatter or bounces). These noise-driven crossings trigger redundant asynchronous processing cycles that do not encode useful signal information. This increases the effective bubble rate and inflates $P_{dynamic}$ well beyond the ideal theoretical prediction. This simultaneous increase in power consumption and reduction in $ENOB$ leads to a severe degradation of the Walden Figure of Merit (FoM_W), highlighting why minimizing noise-induced bubble rates is a critical design constraint for maintaining energy efficiency.

3.5.3 Chapter Summary

This chapter established the theoretical and architectural foundations of the proposed asynchronous Flash ADC, highlighting the paradigm shift from rigid, clock-bound sampling to signal-dependent, data-driven quantization. By eliminating the global clock, the converter addresses the inherent energy inefficiencies typically observed in synchronous architectures when processing sparse or bursty real-world signals.

The structural development began with the adaptation of the comparator array, where continuous-time latched comparators were analyzed to ensure robust operation without synchronous clock phases. The core of the architecture—the asynchronous control logic—was modeled to manage the handshaking, threshold crossing detection, and localized timing generation required to guarantee hazard-free state transitions and mitigate metastability.

Finally, the design space was mathematically bounded through analytical modeling of key physical constraints. It was demonstrated that the maximum input frequency is governed not by Nyquist boundaries, but by a strict Slew Rate constraint (SR_{ADC}), where circuit propagation delays (τ_{conv}) set the limit to prevent slope overload. Furthermore, the power analysis verified that while dynamic power consumption scales exponentially with nominal resolution (N), it retains a highly beneficial linear dependency on the input signal frequency (f_{in}). This analytical framework also illuminated the critical impact of transient noise, which degrades the $SNR/SNDR$, introduces unwanted bubble errors, and triggers redundant conversion cycles that inflate power consumption—ultimately compromising the converter's effective resolution ($ENOB$) and overall energy efficiency (FoM_W). These mathematical insights serve as the direct baseline for the simulation setup and performance evaluation presented in the subsequent chapters.

Chapter 4

Synchronous Reference Flash ADC

4.1 Motivation for Architectural Selection

Benchmarking a novel converter architecture requires a reference point that is close enough structurally to make the comparison genuinely informative. The synchronous reference used in this work is a 4-bit Flash ADC with rail-to-rail comparators, originally presented in [**flashadc_ref**] and designed in a 65-nm CMOS process. For the purpose of this comparison, the architecture was re-implemented and simulated in a 16-nm technology node, under the same conditions applied to the proposed asynchronous converter throughout this dissertation.

The choice of this particular design was not arbitrary. Several properties directly align with those of the asynchronous architecture described in Chapter **??**, and it is precisely this alignment that makes the comparison meaningful. Both converters use the Flash topology, so the number of comparators and the fundamental conversion principle are shared. Both are designed for 4-bit resolution, keeping the complexity of the comparison logic identical. Both feature rail-to-rail input comparators, ensuring that the input voltage swing does not introduce any asymmetry between the two designs. Finally, both architectures are explicitly oriented towards low-power operation, which is the design space where the asynchronous approach is expected to have the most significant impact.

The fundamental difference, and the one this comparison is intended to illuminate, is the presence of a global clock. The synchronous reference depends on a periodic clock to coordinate its comparators and define valid output instants, while the asynchronous architecture eliminates the clock entirely. Removing the clock distribution network is one of the primary motivations for adopting asynchronous operation, since the clock tree contributes a non-trivial fraction of total power in high-speed designs. By holding all other parameters constant and changing only this aspect, the comparison provides a direct and controlled view of what asynchronous operation actually costs or saves in power and performance.

4.2 Architecture Overview

A Flash ADC resolves an analogue input voltage in a single conversion step by comparing it simultaneously against $2^N - 1$ evenly spaced reference levels, where N is the number of output bits. For a 4-bit converter, this means 15 comparators operate in parallel at every active clock edge, each producing a binary decision about whether the input exceeds its assigned threshold. The outputs of all 15 comparators form a thermometer code a sequence of ones followed by zeros with no transitions other than the boundary between them which is then passed to an encoder that maps it to the final 4-bit binary word.

[Insert: top-level block diagram of the synchronous Flash ADC]

Figure 4.1: Top-level block diagram of the synchronous 4-bit Flash ADC [**flashadc_ref**].

The reference voltages fed to each comparator are generated by a resistor ladder connected between V_{REF+} and V_{REF-}, which in this rail-to-rail implementation are tied directly to the supply rails V_{DD} and V_{SS}. The ladder divides the full input range into 15 equally spaced steps of one LSB each, so the k-th comparator from the bottom receives a reference of

$$V_{REF,k} = V_{SS} + k \cdot \frac{V_{DD} - V_{SS}}{2^N}, \qquad k = 1, 2, \ldots, 2^N - 1. \tag{4.1}$$

Since all comparators are driven by the same clock, every decision is made within a single clock period and the output is valid only at well-defined time instants. This is the defining characteristic of a synchronous converter and stands in direct contrast to the asynchronous operation discussed in Chapter **??**.

4.3 The StrongARM Latch Comparator

4.3.1 Circuit Description and Operating Principle

The comparator used in this architecture is the StrongARM latch, a dynamic comparator topology that has become something of a standard reference in both high-speed receiver design and data converter circuits [**razavi_strongarm**]. Its widespread adoption reflects a genuinely attractive combination of characteristics: high regeneration speed, good noise performance, and essentially zero static power dissipation all of which matter in a Flash ADC where 15 comparators must operate in parallel.

[Insert: StrongARM latch schematic]

Figure 4.2: Schematic of the StrongARM latch comparator [**flashadc_ref**].

The circuit operates in two alternating phases controlled by the clock signal a reset phase and an evaluation phase.

The reset phase occurs when the clock is low, the tail transistor is turned off and a pair of reset switches pull both output nodes up to V_{DD}. This precharges the outputs to a known state before every comparison, so that each evaluation begins from the same initial condition regardless of the previous result. Because the tail transistor is off and no current path exists through the core of the circuit, static power dissipation is identically zero during this phase.

The evaluation phase occurs when the clock rises, the tail transistor turns on and current begins to flow through the differential input pair. The input transistors steer this current according to the voltage difference $\Delta V = V_{INP} - V_{INN}$: the side with the higher input voltage conducts more, causing that output node to discharge faster than the other. Once a sufficient imbalance develops across the internal nodes, the cross-coupled inverter pair at the output enters regeneration a positive feedback mechanism that amplifies the initial imbalance exponentially until the outputs settle at valid logic levels [**razavi_strongarm**]. The time needed to complete the regeneration is inversely related to the input overdrive: larger differences between the input voltage and the reference threshold resolve faster, while inputs very close to the threshold require more time. This input-dependent latency is a fundamental property of dynamic comparators and is one of the key differences exploited by the asynchronous architecture.

Since current only flows during the evaluation phase, and only briefly at that, the average power consumption is directly proportional to the clock frequency. This makes the StrongARM latch well suited to low-power converter design: reducing the sampling rate reduces the power linearly.

4.3.2 Rail-to-Rail Input Range

A key property of this comparator implementation is its ability to correctly resolve input voltages across the full supply range, from V_{SS} to V_{DD}. In a Flash ADC, the comparators assigned to the top and bottom reference levels must operate very close to the supply rails, and a comparator that loses sensitivity or gain in those regions would introduce nonlinearity at the extremes of the transfer characteristic, degrading the integral linearity of the converter. The rail-to-rail design ensures this does not happen, maintaining consistent performance across the entire input range.

This property is shared by the asynchronous architecture presented in this work, so it does not introduce any asymmetry into the comparison between the two designs.

4.4 Clock Generation

Since the architecture is synchronous, a stable clock signal must drive the comparator array and determine the conversion instants. For simulation purposes, the clock was generated using a ring oscillator, which provides a compact and realistic on-chip clock source without requiring an ideal external generator.

Rather than driving the comparators directly from the oscillator output, the signal was routed through a tapered inverter buffer chain designed to model the loading and parasitics of a realistic clock distribution network. The chain consists of a sequence of inverters with progressively increasing drive strength, with each stage sized at approximately twice the fanout of the previous one. The final stage of the chain is a $\times 24$ fanout inverter, whose size reflects the capacitive load that a clock signal would encounter when distributed through the metal routing to the ADC core the interconnect wires, vias, and comparator gate inputs that the clock must charge and discharge on every cycle.

A $\times 24$ final buffer is a reasonable and representative choice for modelling the routing from a clock source to the ADC, as it captures the kind of loading one would expect in a physically compact but realistic clock distribution. Neglecting this stage would result in a power estimate that is overly optimistic: in a real implementation, the clock tree is rarely a negligible contributor to total power, and any fair assessment of the synchronous architecture must account for it.

The complete clock path used in simulation is therefore: ring oscillator buffer chain ($\times 2$ fanout per stage) $\times 24$ output buffer comparator array.

4.5 Power Measurement Methodology

To ensure the fairest possible comparison with the proposed asynchronous architecture, the power figures reported in this chapter cover only the core analogue conversion path. Specifically, the encoder the digital logic that translates the 15-bit thermometer code output of the comparators into the final 4-bit binary word was excluded from all power measurements.

The reasoning is straightforward, the asynchronous architecture presented in this work also does not include encoder power in its reported figures. Including it on one side of the comparison but not the other would artificially inflate the synchronous power budget and lead to a conclusion that does not fairly reflect the underlying architectural trade-off. Since both designs require some form of thermometer-to-binary conversion at their outputs, and since this encoding logic is largely independent of whether the preceding conversion is synchronous or asynchronous, the most defensible approach is to exclude it from both and compare only the converter cores.

The power reported in Table 4.1 therefore includes the resistor ladder, the 15 StrongARM latch comparators, the ring oscillator, and the tapered clock buffer chain described in the previous section.

4.6 Simulation Results

Table 4.1: Simulated performance summary of the synchronous reference Flash ADC in 16-nm CMOS.

Parameter	**Value**	**Unit**
Supply Voltage		V
Resolution		bit
Sampling Frequency		MS/s
Power Consumption		μW
Rail-to-Rail Input	Yes	–
Bubble Rate		%
FoM_W (Walden)		fJ/conv.-step
Dynamic Performance (without Noise)		
SNR		dB
SINAD / SNDR		dB
ENOB		bit
Dynamic Performance (with Transient Noise)		
SNR		dB
SINAD / SNDR		dB
ENOB		bit

Chapter 5

Trimming

5.1 Body Biasing Trimming

5.1.1 Function of NMOS and PMOS Transistors Under Different Body Bias

The threshold voltage of a MOSFET is not solely determined by the device geometry and doping profile but is also a function of the potential applied to its body terminal relative to the source. This dependence, commonly referred to as the body effect or substrate bias effect, is described for an NMOS transistor by

$$V_{tn} = V_{tn0} + \gamma_n \left(\sqrt{2\phi_{F,n} + V_{SB}} - \sqrt{2\phi_{F,n}} \right), \tag{5.1}$$

where V_{tn0} is the zero-bias threshold voltage, γ_n is the body effect coefficient, $\phi_{F,n}$ is the Fermi potential, and V_{SB} is the source-to-body voltage. The analogous expression for a PMOS transistor, accounting for the sign conventions appropriate to a p-channel device, takes the form

$$|V_{tp}| = |V_{tp0}| + \gamma_p \left(\sqrt{2\phi_{F,p} + |V_{SB}|} - \sqrt{2\phi_{F,p}} \right), \tag{5.2}$$

where the relevant source-to-body voltage for the PMOS is determined by the potential difference between the source and the N-well. In both cases, an increase in $|V_{SB}|$ raises the magnitude of the threshold voltage, a condition referred to as reverse body biasing (RBB) conversely, reducing $|V_{SB}|$ below its nominal value or, in technologies that permit it, forward-biasing the body junction lowers $|V_t|$, a condition known as forward body biasing (FBB).

A direct consequence for the slicer inverters in the proposed architecture is that each cell's trip point, given by equation (3.4), depends on the threshold voltages of both transistors composing the inverter. By substituting the body-bias-dependent expressions for V_{tn} and $|V_{tp}|$ into the inverter threshold equation and expanding to first order, it is apparent that a positive shift in V_{tn} achieved by applying RBB to the NMOS weakens the pull-down network relative to the pull-up network and therefore raises V_M, while an equivalent increase in $|V_{tp}|$ weakens the pull-up network and lowers V_M. This asymmetry between the NMOS and PMOS body bias responses provides two independent degrees of freedom for adjusting the trip point of each slicer: the NMOS bulk voltage controls the effective strength of the pull-down path, while the PMOS bulk voltage (equivalently, the N-well potential) controls the pull-up path. Applying corrections simultaneously to both body terminals therefore allows independent tuning of the trip point magnitude and of the symmetry of the transition characteristic, providing a richer correction space than either terminal alone could offer.

5.1.2 Process Corner Dependence and Motivation for Trimming

In a deep-submicron CMOS process, fabrication variability causes the threshold voltages and carrier mobilities of the NMOS and PMOS devices to deviate from their nominal (typical-typical, TT) values. The standard process corners used to bound this variability slow-slow (SS), fast-fast (FF), fast-NMOS slow-PMOS (FS), and slow-NMOS fast-PMOS (SF) capture the four extreme combinations of NMOS and PMOS behaviour and represent the conditions under which circuit performance is most likely to degrade.

For the slicer-based thermometer coder described in the previous chapter, the impact of process variation is particularly significant because the thermometer code relies on the precise relative ordering of the inverter trip points across the output bit bank. At the typical corner, the ratios r_k assigned to each slicer place the trip points at the intended positions relative to the reference level of 0.4 V, and the resulting differential non-linearity (DNL) and integral non-linearity (INL) of the converter remain within specification. At non-typical corners, however, the effective transistor strengths deviate from their nominal values in a manner that is not uniform across bits: since the PMOS and NMOS devices within each slicer are affected differently depending on the corner, the trip point of each inverter shifts by a bit-dependent amount, breaking the intended monotonic spacing and introducing non-linearities that manifest as elevated DNL and INL.

At the SS corner, both NMOS and PMOS threshold voltages increase, reducing the drive strength of both pull-down and pull-up networks. Although the symmetric nature of this degradation might suggest a self-compensating effect on the trip point, the body effect coefficients γ_n and γ_p and the nominal threshold values are in general not matched between the two device types, so the net shift in V_M is non-zero and, more importantly, bit-dependent due to the different sizing ratios across the slicer bank. The FF corner produces the complementary effect: reduced threshold voltages increase the drive strength of both device types, again shifting the trip points by amounts that vary across the bit cells. The FS and SF corners are qualitatively different in that they introduce an asymmetric perturbation: at FS, the NMOS is faster than nominal while the PMOS is slower, biasing all slicers toward a lower trip point and compressing the effective decision levels toward the bottom of the input range at SF, the complementary imbalance expands the decision levels toward the top. In all four non-typical corners, the degradation in linearity was found to cause DNL excursions in excess of 0.5 LSB, rendering the converter non-monotonic under these conditions.

5.1.3 Offline Trimming Strategy via Body Bias Adjustment

To restore the linearity of the thermometer coder across process corners without modifying the core circuit topology, an offline trimming strategy based on body bias adjustment was adopted. The principle of the approach is to apply distinct bulk voltages to the NMOS and PMOS transistors of the output slicer stage independently for each device type in a manner calibrated to shift the trip points back toward their nominal positions. Since both the NMOS bulk voltage V_{BN} and the PMOS bulk voltage V_{BP} contribute independently to the trip point through equations (5.1) and (5.2), respectively, adjusting both simultaneously allows correction of both the common-mode shift in trip point which affects DNL and the differential imbalance between NMOS-dominated and PMOS-dominated corners which affects INL. The trimming is defined as offline in the sense that the bulk voltages are determined once, based on the identified process corner, and held constant during normal operation no continuous adaptive feedback is involved.

For each of the four non-typical corners, a specific pair of values (V_{BN}, V_{BP}) was determined such that the DNL of the converted thermometer code falls below 0.5 LSB across the full input range. At the SS corner, where both device types exhibit elevated threshold voltages and therefore reduced drive strength, forward body biasing is applied to both the NMOS and PMOS, reducing V_{tn} and $|V_{tp}|$ toward their nominal values and thereby restoring the intended trip point positions. At the FF corner, reverse body biasing is applied to both types to counteract the excessive drive strength resulting from the low threshold voltages. The FS and SF corners require asymmetric corrections: at FS, a stronger reverse bias is applied to the NMOS bulk, weakening the pull-down network to compensate for the fast-NMOS tendency, while the PMOS bulk is forward-biased to partially recover the reduced pull-up strength the SF corner is handled symmetrically, with the dominant correction applied to the PMOS bulk.

The trimming was implemented in simulation using a Verilog-A behavioural model. The model monitors the process corner identifier and, based on the detected corner, drives V_{BN} and V_{BP} to the pre-computed correction values. This approach allows the trimming to be evaluated across all corners within the same simulation framework as the nominal design, without requiring modifications to the transistor-level netlist. The Verilog-A abstraction also provides a clean interface for extending the trimming table to intermediate corner points or to temperature-corner combinations should a finer correction granularity be required in future iterations. The effect of the body bias corrections on the DNL and INL of the converter is presented in Chapter **??**.

Chapter 6

Simulation Results and Performance Benchmarking

6.1 Simulation Setup and Input Test Signals

To validate the theoretical framework established in Chapter 3 and evaluate the physical performance of the proposed asynchronous Flash ADC, extensive transistor-level simulations were conducted using a 16-nm CMOS process node under a 0.8 V nominal supply voltage. This section details the simulation environment, the specialized reconstruction methodology required for non-uniform asynchronous data, and the characteristics of the input test stimuli.

Unlike conventional synchronous data converters where output samples are bound to a periodic clock grid, the asynchronous level-crossing Flash ADC generates discrete output tokens only when a quantization threshold is traversed. Consequently, traditional Fast Fourier Transform (FFT) algorithms, which strictly require uniformly sampled time-domain records, cannot be directly applied to the raw asynchronous output. To accurately characterize the dynamic performance metrics such as Signal-to-Noise Ratio (SNR), Signal-to-Noise-and-Distortion Ratio (SNDR), and Effective Number of Bits (ENOB) a non-uniform to uniform time-domain resampling and mathematical reconstruction algorithm was implemented.

The primary input test signal employed for dynamic characterization is a full-scale sinusoidal waveform near the Nyquist limit, superimposed with transient thermal noise to model realistic operating conditions. The spectrum of the reconstructed asynchronous output, showcasing the fundamental signal bin, the noise floor, and harmonic distortion components, is illustrated in the FFT spectrum below.

The dynamic performance curves, mapping the behavior of both SNDR and ENOB across a broad range of input frequencies and amplitudes, are provided in Figure 6.2. These curves contrast the ideal noiseless performance against the degraded parameters under transient noise conditions, tracking the incidence of noise-induced threshold chatter.

Figure 6.1: FFT spectrum of the reconstructed asynchronous Flash ADC output signal.

Figure 6.2: SNDR and ENOB as a function of the input signal frequency and amplitude.

6.2 Static Linearity Results

Static linearity metrics, specifically Differential Non-Linearity (DNL) and Integral Non-Linearity (INL), were evaluated to quantify the architectural and circuit-level deterministic mismatches. In an asynchronous Flash ADC, DNL and INL are heavily influenced by the random threshold variations of the comparator array and the parasitic resistance profile of the reference ladder.

To counteract these non-idealities, a body bias trimming technique was proposed and implemented via a Verilog-A behavioral framework, as described in Chapter 5. To demonstrate the efficacy of this calibration mechanism, the static error profiles were extracted before and after the application of the optimal trimming vectors across multiple process corners.

Figure 6.3 illustrates the untrimmed versus trimmed DNL and INL profiles across the converter's full-scale digital codes, highlighting the reduction in peak errors and the elimination of potential missing codes.

Figure 6.3: Measured DNL and INL error profiles before (top) and after (bottom) the application of the body bias trimming calibration.

The summary of the peak and root-mean-square (RMS) static errors extracted from the simulation runs is presented in Table 6.1.

Table 6.1: Static linearity performance (DNL and INL) summary before and after calibration.

	Before Trimming		**After Trimming**	
Metric (LSB)	**Peak**	**RMS**	**Peak**	**RMS**
DNL				
INL				

6.3 Power Dissipation and Energy Efficiency

A defining feature of the asynchronous Flash ADC is its signal-dependent power profile. In direct contrast to traditional synchronous architectures that exhibit a static power baseline due to continuous clock toggling, the dynamic power of the proposed converter is governed strictly by the frequency of threshold-crossing events.

To empirically validate this adaptive scaling behavior, the total power consumption was monitored across an input signal frequency sweep spanning from DC ($f_{in} = 0$) up to the maximum tracking bandwidth limit. Figure 6.4 explicitly demonstrates this linear dependency, proving that the system enters a near-zero dynamic consumption state when the input stimulus exhibits low activity or becomes idle.

Figure 6.4: Total and dynamic power consumption as a function of the input signal frequency (f_{in}).

To provide a mathematically rigorous assessment of energy efficiency in this non-uniform context, an adapted Walden Figure of Merit ($FoM_{\mathrm{W},adapted}$) was utilized. This metric incorporates the effective average sampling rate ($f_{s,avg}$) dictated by the crossing density, ensuring a fair benchmark against synchronous designs:

$$\mathrm{FoM}_{W,adapted} = \frac{P_{total}}{2^{\mathrm{ENOB}} \cdot f_{s,avg}} \tag{6.1}$$

The extracted power breakdown and the resulting energy efficiency figures under both ideal and transient noise conditions are tabulated in Table 6.2.

6.4 Comparative Analysis and Discussion

To contextualize the performance of the developed design, a comprehensive comparative analysis is carried out. Table 6.3 contrasts the core performance metrics of the baseline Synchronous Reference Flash ADC detailed in Chapter 4, the proposed Asynchronous Flash ADC developed in this work, and recent state-of-the-art architectures compiled from the literature.

The comparative data underscores the trade-offs inherent to the asynchronous paradigm. While the synchronous baseline maintains a lower circuit complexity in its control block, it suffers from

Table 6.2: Power dissipation breakdown and energy efficiency summary.

Performance Parameter	**Value**	**Unit**
Static Power Consumption (P_{static})		μW
Dynamic Power (w/o Noise)		μW
Dynamic Power (with Transient Noise)		μW
Total Power Dissipation		μW
FoM$_W$ (w/o Noise)		fJ/conv.-step
FoM$_W$ (with Transient Noise)		fJ/conv.-step

Table 6.3: Performance comparison with the synchronous baseline and state-of-the-art literature.

Parameter	**Synchronous Baseline (Chapter 4)**	**This Work (Asynchronous)**	**Ref. [X]**	**Ref. [Y]**	**Ref. [Z]**
Technology (nm)	16	16			
Supply Voltage (V)	0.8	0.8			
Resolution (Bits)					
f_s or $f_{s,avg}$ (MS/s)					
SNDR (dB)					
ENOB (Bits)					
Power (μW)					
FoM$_W$ (fJ/c.-step)					
Architecture	Synchronous Flash	Asynchronous Flash			

a rigid power envelope that compromises efficiency during periods of low input activity. In contrast, the proposed asynchronous implementation achieves superior energy scaling and a highly competitive Walden Figure of Merit, particularly after the activation of the body bias trimming network which successfully recovers the design's effective resolution under process variations.

Chapter 7

Conclusions and Suggestions for Future Work

7.1 Summary of the Work Developed

This dissertation addressed the design and validation of an asynchronous Flash Analog-to-Digital Converter intended for low-power, event-driven applications such as Internet of Things sensor nodes and biomedical signal acquisition interfaces. The central motivation for this work stems from the fundamental inefficiency of conventional synchronous ADC architectures when dealing with time-sparse signals: a global clock forces continuous switching activity across the comparator bank and clock distribution network, dissipating dynamic power regardless of whether the input signal is actually carrying new information. Removing that clock and replacing it with a level-crossing, event-driven mechanism was the core hypothesis explored throughout the dissertation.

The work began with a thorough review of the state of the art in data conversion. Classical ADC architectures Flash, SAR, Pipeline, and Sigma-Delta were examined with particular attention to the trade-offs between speed, resolution, and power consumption. This analysis was grounded in performance data from 140 publications drawn from ISSCC and JSSC, from which Walden and Schreier figures of merit were extracted and compared across a wide range of sampling frequencies and technology nodes. The review made clear that synchronous Flash converters occupy the high-speed frontier but at the cost of a poor energy figure of merit, and that asynchronous, event-driven operation is one of the most promising paths toward breaking this constraint for sparse-signal applications. A dedicated discussion on the adaptation of standard figures of merit to asynchronous systems was also included, since applying a fixed Nyquist sampling frequency to a level-crossing converter is inherently inappropriate and leads to misleading efficiency estimates.

The design of the asynchronous ADC itself evolved through three distinct architectural phases. The first candidate was a current-mirror flash configuration in which a transconductance cell converts the differential input voltage into an error current, and a current-to-code converter maps that current to a thermometer-coded output by exploiting asymmetric PMOS and NMOS mirror ratios at each comparison node. This topology operates in continuous time and requires no external clock, making it a natural fit for the event-driven paradigm. The second phase investigated a complementary dual-input architecture aimed at extending the input common-mode range toward rail-to-rail operation by combining an NMOS and a PMOS differential pair with cross-coupled output branches. Although conceptually attractive, this approach could not be made to work within the 0.8 V supply budget of the target 16 nm process: the stacking of additional transistors in the signal path exceeded the available voltage headroom, and no feasible bias point could be found that kept all devices in saturation simultaneously. The third and final architecture abandoned the rail-to-rail extension and returned to a five-transistor OTA configured as a unity-gain voltage follower, whose low-impedance output drives a bank of CMOS slicer inverters. Each slicer is sized with a distinct PMOS-to-NMOS transistor ratio, which sets its switching threshold at a specific voltage level within the supply range. The ensemble of slicers produces an asynchronous thermometer code whose transitions directly encode level-crossing events, with no clock required at any point in the signal path.

Alongside the asynchronous core, a synchronous reference Flash ADC was implemented in the same 16 nm technology and under the same supply conditions to serve as a fair benchmark. This reference uses fifteen StrongARM latch comparators driven by a ring oscillator with a tapered buffer chain, and its power budget explicitly accounts for the clock distribution network, which is precisely the overhead that the asynchronous design aims to eliminate. The comparison between the two converters therefore isolates the effect of removing the global clock from an otherwise structurally similar Flash architecture.

A critical practical challenge of the slicer-based thermometer coder is its sensitivity to process variation. Since the threshold of each inverter is determined by the ratio of PMOS to NMOS drive strength, any departure from the nominal device parameters shifts the trip points in a bit-dependent manner, degrading the differential and integral non-linearity of the converter. To address this, an offline body bias trimming strategy was designed and implemented using a Verilog-A behavioural model. By applying distinct bulk voltages to the NMOS and PMOS transistors of the slicer stage, the threshold voltage of each device type can be shifted through the body effect, allowing the trip points to be steered back toward their nominal positions after fabrication. Correction vectors were determined for the four standard process corners SS, FF, FS, and SF and were shown in simulation to reduce the peak DNL and INL errors to within specification.

Finally, the complete system was validated through transistor-level analog and mixed-signal co-simulation. The dynamic performance of the asynchronous converter was characterised using a non-uniform to uniform resampling algorithm applied to the raw level-crossing output, enabling standard spectral metrics such as SNDR and ENOB to be extracted and compared against both the synchronous reference and representative state-of-the-art designs. The power measurements confirmed the key theoretical prediction: dynamic power consumption scales linearly with input signal frequency, and the converter enters a near-zero dynamic consumption state when the input is idle. The adapted Walden figure of merit, computed using the effective average crossing rate rather than a fixed Nyquist frequency, demonstrates a clear energy efficiency advantage over the synchronous baseline for the sparse-signal operating regime targeted by this work.

7.2 Concluded Objectives

The four specific objectives defined at the outset of this dissertation were all successfully accomplished.

The first objective was the design of an asynchronous Flash ADC architecture that dissipates dynamic power only when input signal events occur. This was achieved through the final five-transistor OTA plus ratioed-slicer topology, which operates in continuous time without any global clock signal. The theoretical analysis confirmed that the converter's dynamic power scales linearly with the input signal frequency and approaches zero during idle intervals, directly validating the intended activity-dependent power profile.

The second objective was the implementation of an offline trimming mechanism capable of detecting and compensating the threshold voltage mismatch of the comparators across process corners. This was fulfilled through the body bias trimming strategy described in Chapter 5, in which pre-computed pairs of NMOS and PMOS bulk voltages are applied to the slicer bank to restore the intended trip-point positions. The approach was validated across the SS, FF, FS, and SF process corners, achieving a significant reduction in both peak and RMS DNL and INL errors, as reported in Table 6.1.

The third objective was the validation of the complete system through analog and digital co-simulation. The transistor-level netlist of the asynchronous ADC was simulated in a 16 nm CMOS process under a 0.8 V supply, and both static linearity and dynamic performance metrics were extracted. The FFT-based spectral analysis, performed on the uniformly resampled level-crossing output, confirmed correct operation of the asynchronous core and the functionality of the trimming logic, satisfying this objective.

The fourth objective was performance benchmarking against synchronous state-of-the-art designs using established figures of merit. The proposed converter was compared against the synchronous reference Flash ADC implemented under identical technology and supply conditions, as well as against relevant designs from the ISSCC and JSSC literature. The adapted Walden figure of merit confirms that the asynchronous architecture achieves superior energy efficiency for sparse-signal operating conditions, closing a meaningful portion of the power gap that synchronous Flash converters exhibit at high sampling rates.

It should be noted that, although process corner variation was successfully addressed by the offline trimming strategy, Monte Carlo statistical variation was not corrected within the scope of this work. The body bias trimming vectors were determined for the four deterministic process corners and are therefore not adapted to the continuous distribution of device mismatch that Monte Carlo analysis exposes. This limitation is acknowledged as an important gap between the presented prototype and a fully robust production-ready implementation, and it is the primary motivation for the first suggestion for future work described in the following section.

7.3 Future Work

Two directions for future development are identified as the most impactful extensions of the work presented in this dissertation.

7.3.1 Analogue Process Corner Detection Circuit

The trimming strategy validated in this work relies on the correct identification of the active process corner, which in the current implementation is supplied externally to the Verilog-A behavioural model and used to select the appropriate pre-computed correction vector. In a physical integrated circuit this external identification is not available: the fabricated die has no direct means of knowing which combination of NMOS and PMOS device parameters it has been assigned by the foundry.

A natural extension would therefore be to design a dedicated on-chip analogue process detection circuit capable of sensing the effective transistor characteristics of the fabricated device and communicating the inferred process corner directly to the trimming logic. Such a circuit could be implemented as a small reference cell for example, a pair of matched inverter-based ring oscillators, one biased toward NMOS-dominated operation and the other toward PMOS-dominated operation whose relative oscillation frequencies or propagation delays encode the current process corner. The ratio between the two delay measurements maps unambiguously onto the NMOS–PMOS balance of the process, distinguishing SS from FF and FS from SF corners. The resulting analogue measurement would be digitised by a simple comparator and used to address a lookup table of pre-computed body bias correction values, replacing the external corner identification currently provided by the Verilog-A model. This approach would make the trimming procedure fully self-contained and autonomous, removing the need for any external process characterisation while preserving the static, low-overhead nature of the offline correction strategy.

7.3.2 Monte Carlo Trimming

A more fundamental limitation of the current trimming approach is that it corrects deterministic inter-corner shifts but leaves intra-die stochastic mismatch unaddressed. In practice, even at the nominal process corner, individual transistors within the slicer bank will exhibit random threshold voltage variations whose statistical distribution is characterised by Pelgrom's law. These variations are not captured by the four-corner trimming table, and their cumulative effect on DNL and INL can be significant in a Flash converter where each comparator threshold must be individually accurate.

Extending the calibration to handle Monte Carlo variation would require moving from a corner-indexed lookup table to a converter-specific foreground calibration routine capable of measuring the actual trip point of each slicer independently and computing an individual correction for each one. One viable approach would be to inject a known ramp stimulus at the converter input during a dedicated startup calibration phase and record the digital code transitions to infer the actual trip point of each slicer. The deviation of each measured trip point from its nominal target could then be mapped to the body bias correction required to bring it back into specification, using the sensitivity relationship established in Chapter 5. Since the NMOS and PMOS bulk voltages of each slicer would need to be individually controlled, this approach requires the addition of a per-cell bias generator, which increases the area overhead relative to a global corner correction. Nevertheless, the achievable linearity improvement and the corresponding reduction in post-fabrication yield loss would likely justify this added complexity in any application where the converter is produced at volume and must meet tight DNL specifications across the full statistical distribution of process variability.